

Machine Learning in Finance: practice

Tony Guida, Senior Investment Manager – Quant PM

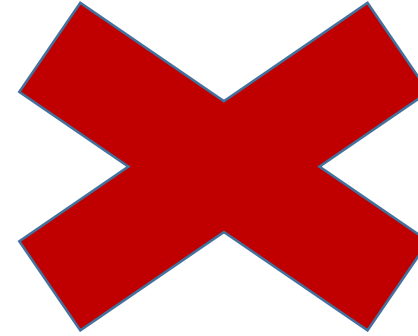
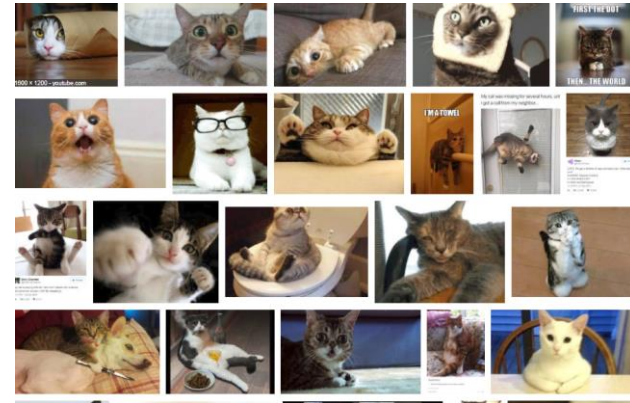
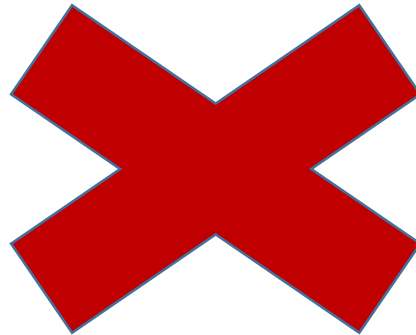
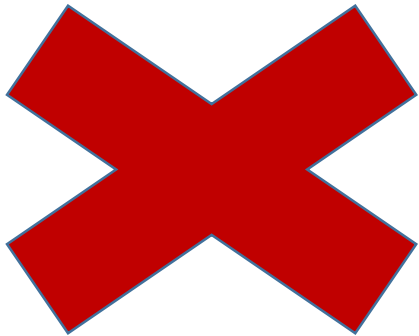


What this presentation is NOT about



#AlphaGo WINS!!!! We landed it on the moon.
So proud of the team!! Respect to the amazing
Lee Sedol too

RETWEETS 246 LIKES 222



Part I: Quant Equity Nowadays, Everybody is a Quant



The search for Quant equity innovation...

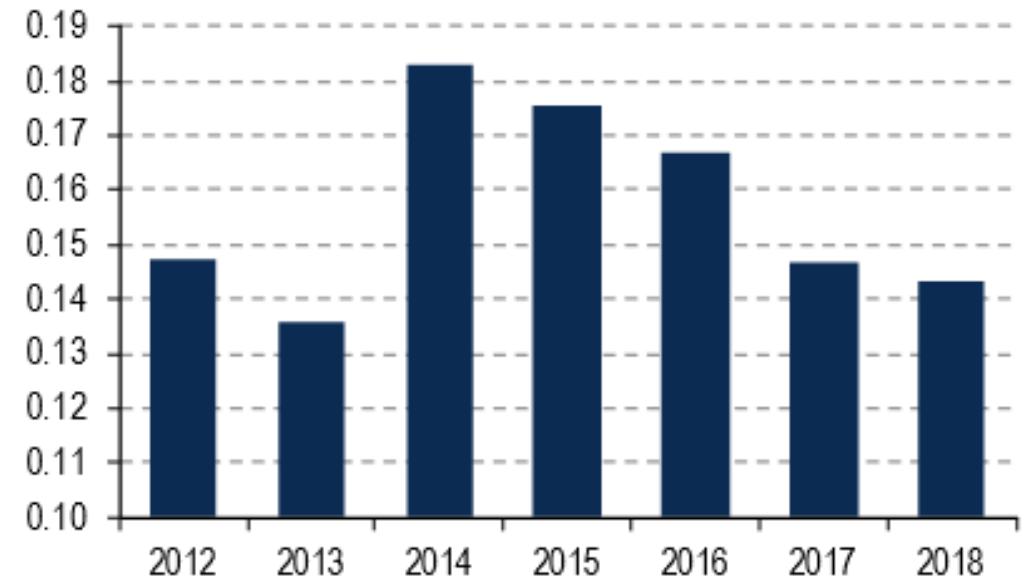
Chart 7: Average number of factors used by respondents, 1989-2018



Note: Does not include data for 2008-10 due to lack of sufficient responses
Source: BofA Merrill Lynch US Equity & US Strategy

Chart 8: Correlations between clients have generally decreased (as clients are using different factors than their peers)

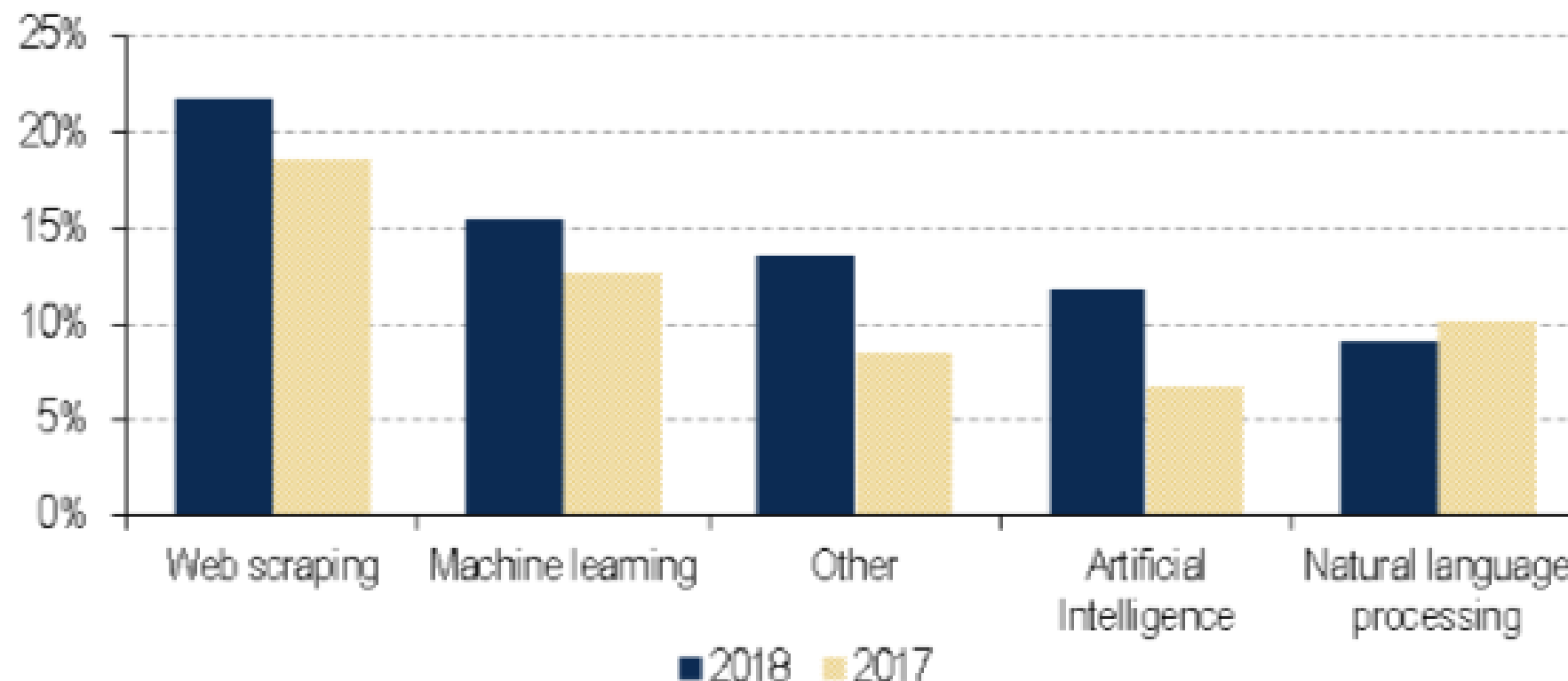
Avg. pair-wise correlations between survey respondents' factor usage by year



Source: BofA Merrill Lynch US Equity & US Strategy

... in Technology

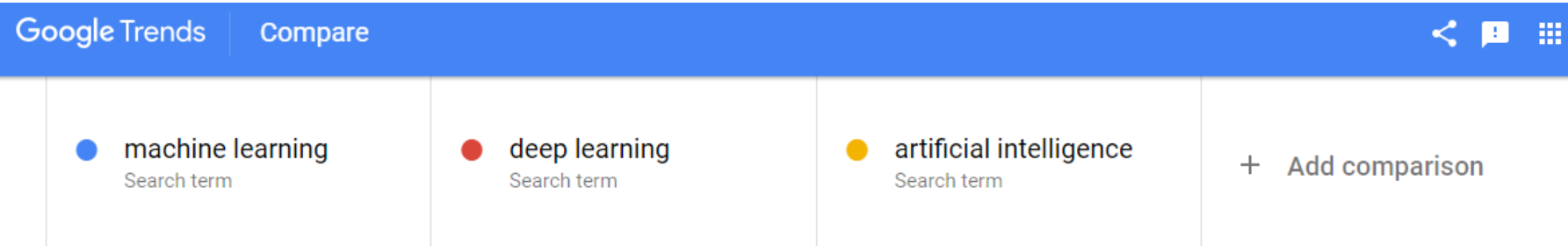
Chart 13: Which of the following quantitative tools do you use in your investment process? (% of 2018 and 2017 survey respondents)



Note: "Natural language processing" answer choice renamed in 2018 survey from "Text-based algorithms" previously

Source: BofA Merrill Lynch US Equity & US Quant Strategy

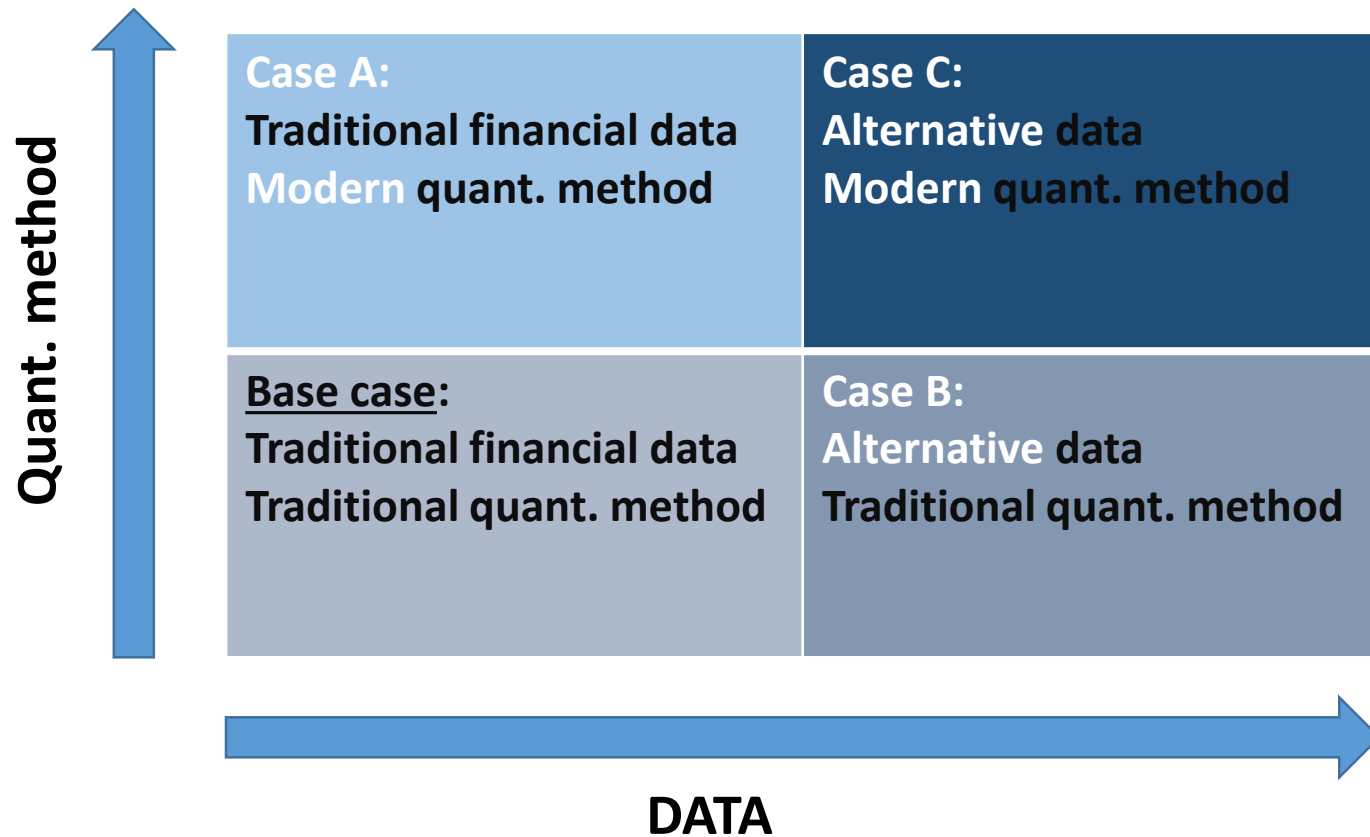
Quant (R)evolution ?



Worldwide 7/9/11 - 8/9/17 Finance Web Search



Quant (R)evolution ?



Usual criticisms against ML in finance

- **Low** signal to noise ratio
- Blackbox, **cannot be interpreted**
- Overfit, **not valid out of sample**
- **Only** for high frequency
- Data mining, prone to **spurious correlation**

Limitations of ML in quantitative investment

- Finance is not a “**plug and play**” for ML
- **Causality** between labels and outcomes
- **Laws** of the **financial ecosystem** change with agents' interactions

Beyond Machine Learning buzzword in finance...

How ?

Intelligence is NOT
consciousness

Technology is NOT
deterministic

- You have to : have a ML “**Why**”
- You have to : **frame** the question, put financial and economic structure in dataset. Your domain knowledge.
- You have to : **understand** and know your **Data**

Modern approaches are made for PREDICTIONS

Journal of Economic Perspectives—Volume 31, Number 2—Spring 2017—Pages 87–106

Machine Learning: An Applied Econometric Approach

Sendhil Mullainathan and Jann Spiess

Machines are increasingly doing “intelligent” things: Facebook recognizes faces in photos, Siri understands voices, and Google translates websites. The fundamental insight behind these breakthroughs is as much statistical as computational. Machine intelligence became possible once researchers stopped approaching intelligence tasks procedurally and began tackling them empirically. Face recognition algorithms, for example, do not consist of hard-wired rules to scan for certain pixel combinations, based on human understanding of what constitutes a face. Instead, these algorithms use a large dataset of photos labeled as having a face or not to estimate a function $f(x)$ that predicts the presence y of a face from pixels x . This similarity to econometrics raises questions: Are these algorithms merely applying standard techniques to novel and large datasets? If there are fundamentally new empirical tools, how do they fit with what we know? As empirical economists, how can we use them?¹

Goals of Prediction and Estimation

- Goal of prediction: loss minimization

$$\hat{f} \approx \min_{f \in \mathcal{F}} E_{(y,x)} L(f(x), x)$$

- Use the data to pick a function that does well on a new data point

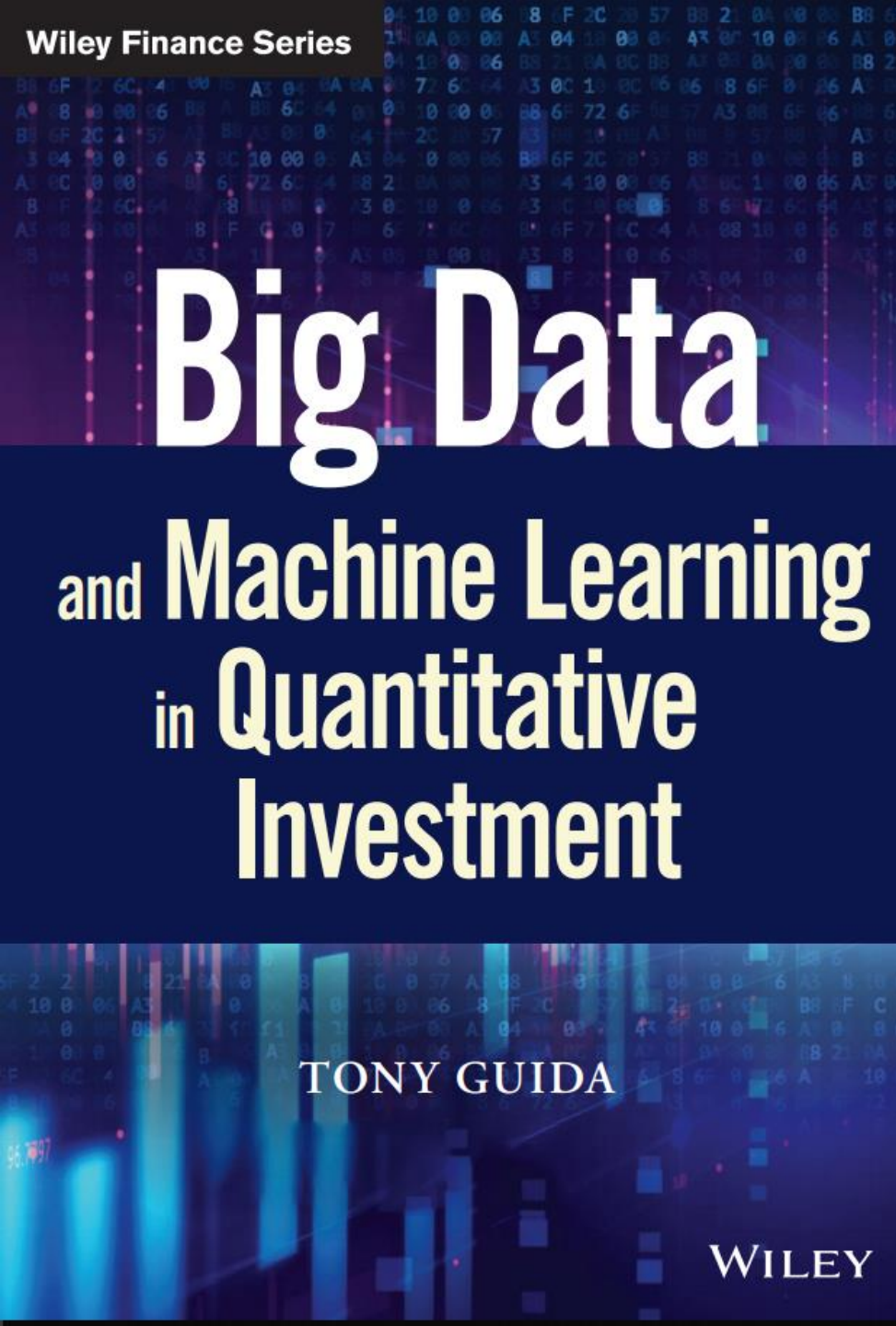
- Goal of estimation: unbiasedness

$$E[\hat{f}] = f$$

Source: AFA Lecture: Machine Learning and Prediction in Economics and Finance

PART II: Big Data and Machine Learning, The book.





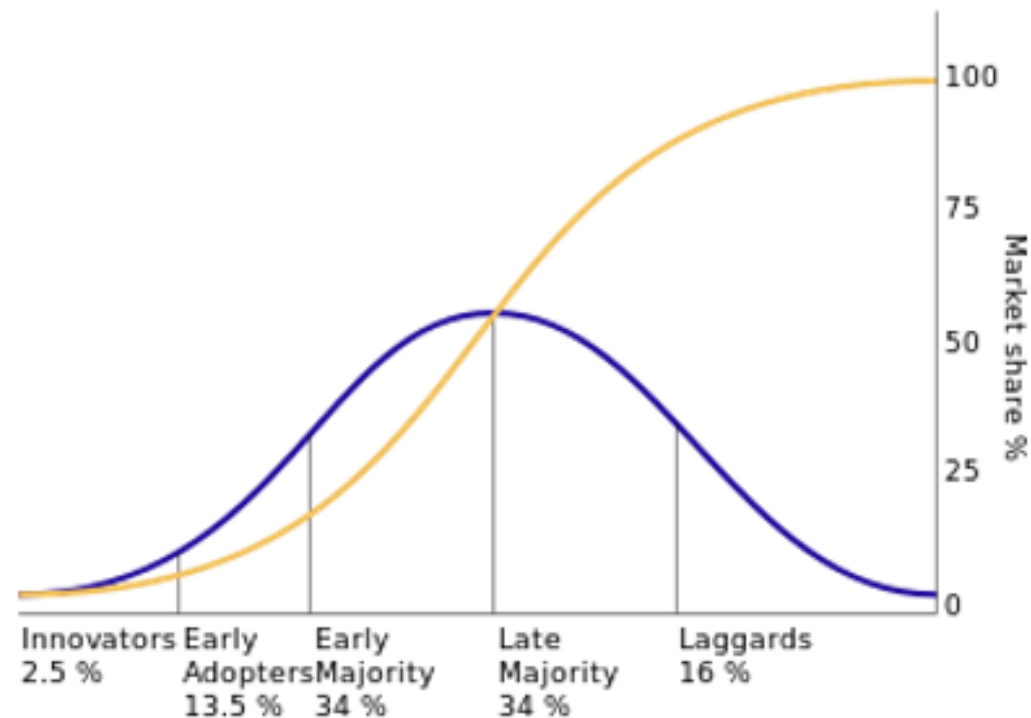
**A book by
practitioners for
practitioners**

Why doing a book ?

- The way Alternative data and ML in finance have been promoted **was wrong**.
- **Wisdom** of the crowd approach.
- **Lack of content** for real application.
- I Believe in the concept of “**quant community**”
- **A need for elaborated debate**

Key takeaways from the book

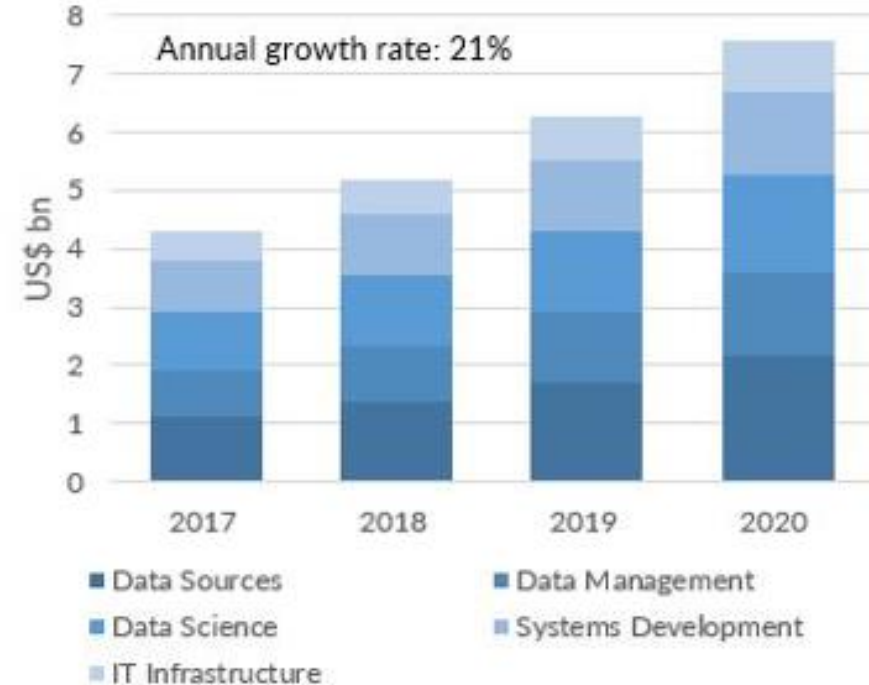
...“The end of the beginning for alternative data”



Source: Rogers Everett - Based on Rogers, E. (1962) *Diffusion of innovations*¹

Source: R. Lipus, Neudata. Chapter 2: Taming Big Data

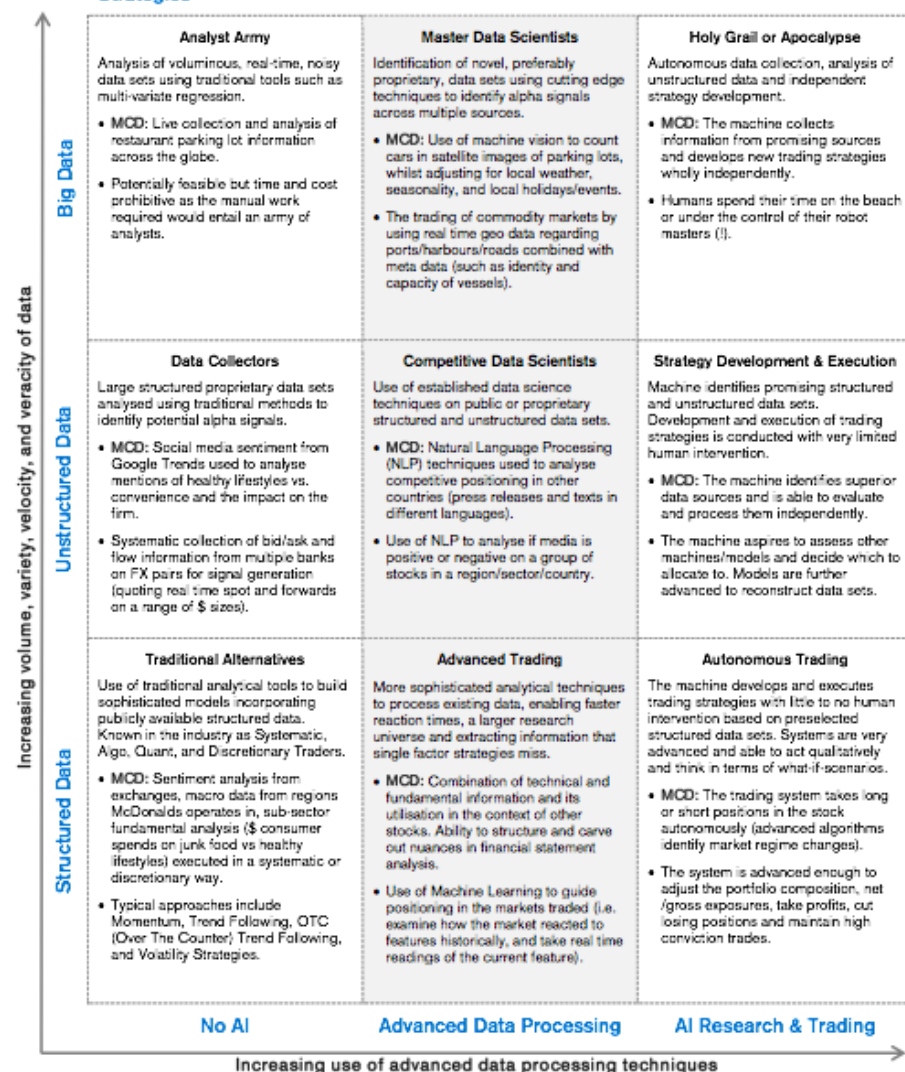
Spending on Alternative Data



Source: Opimas Analysis⁴

Key takeaways from the book

Chart: Applications of Big Data and Artificial Intelligence in Alternative Investment Strategies



Source: E. Sirotyup, CS. Chapter 3: State of machine learning applications in investment management

Key takeaways from the book



"IBM - I'm not leaving any time soon..."
 ★★★★★ Current Employee - Marketing Manager in London, England

Recommendations **Approves of CEO**

Pros
 Terrific breadth of career opportunities - I've worked for IBM since university (12 years now), and I've held 10 different roles, and had opportunity to be part of 3 different start-ups with opportunity for rapid advancement.

IBM also offers great flexible working opportunities - I work a condensed working week now that I have children, and my job is mostly based at my home office.

Cons
 Strong US work ethic - rare for anyone in my department to leave on time in the evening, and overtime is not paid. Really hard to work part-time in a company like IBM - you really need to have a presence 5 days per week unless you're happy for your career to go on the back-burner for a while...
 Maternity leave pay is pretty rubbish, sliding scale down to 40% pay (better than a few years ago, but still poor).

	(1)	(2)	(3)
lagged dependent	-0.012 (-0.358)	-0.015 (-0.423)	-0.012 (-0.351)
Forecast dispersion	-2.700 (-3.196)	-2.806 (-3.318)	-2.581 (-2.916)
OVERALL	0.067 (0.761)	0.053 (0.505)	0.079 (0.755)
GOAL		1.770 (2.536)	4.477 (3.751)
TONE		0.054 (2.071)	1.714 (1.796)
High_expectations			14.180 (2.892)
Analyst revisions	15.130 (4.749)	14.730 (4.622)	18.050 (5.173)
Log(Market Equity)	0.000 (-1.078)	0.000 (-1.021)	0.000 (-1.552)
Log(Book/Market)	-0.006 (-0.096)	-0.018 (-0.294)	-0.053 (-0.857)
Pmom	0.716 (7.411)	0.738 (7.612)	0.774 (8.007)
KLD			0.055 (1.904)
BC			-0.974 (-1.699)

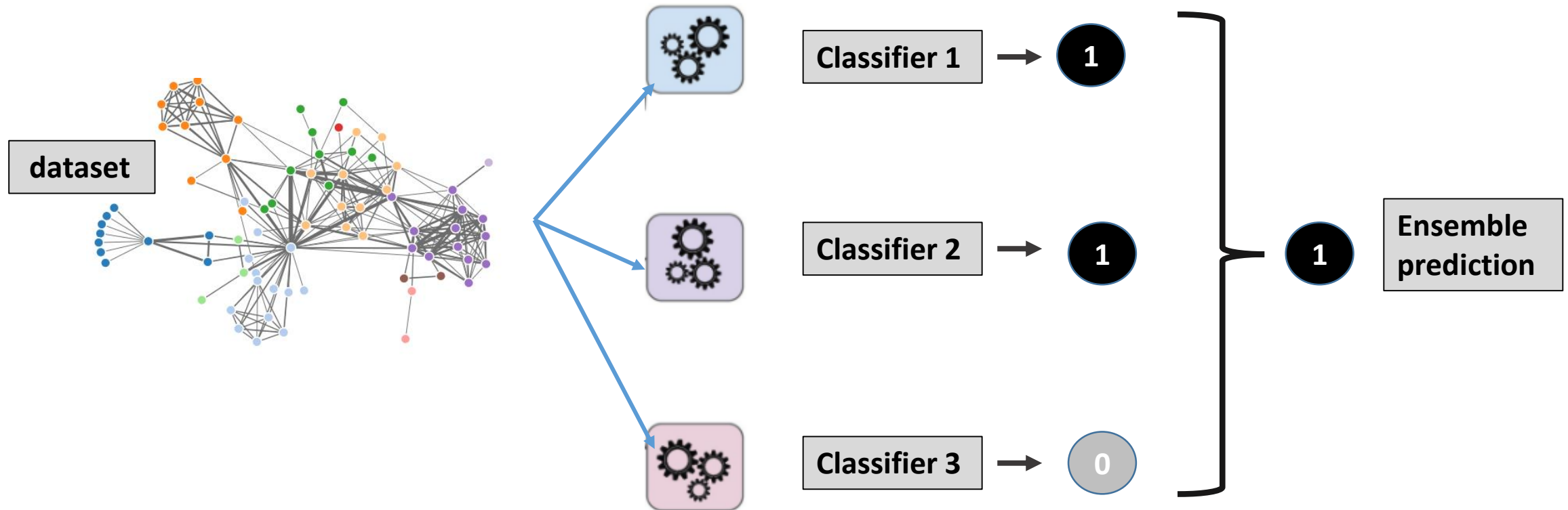
PART III: Training Equity Models



Concepts and definitions

Introduction and definition: Ensemble Learning

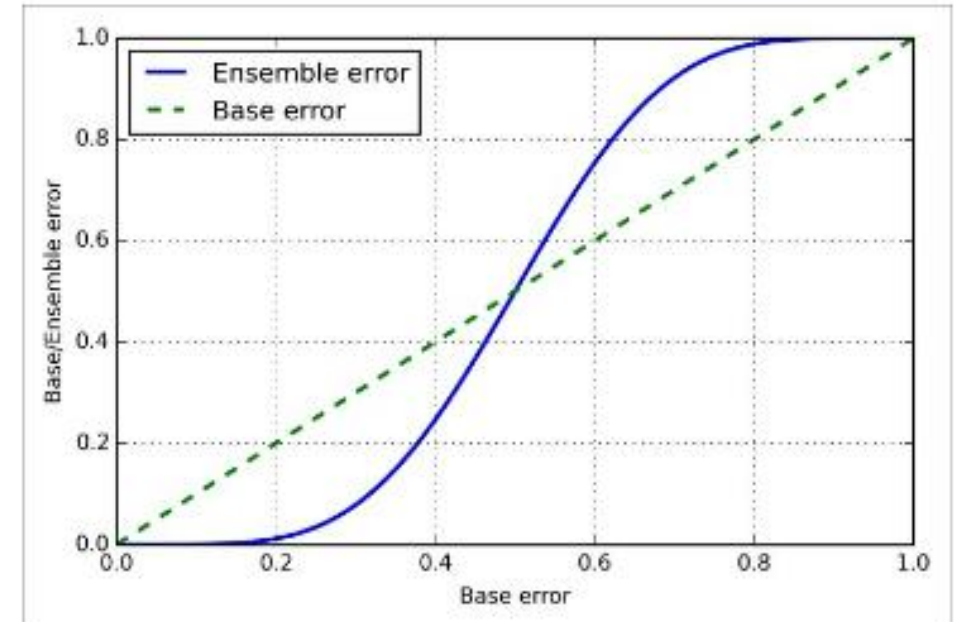
“**Ensemble Learning purpose** is to combine the predictions of several classifiers built with a given learning algorithm in order to improve bias and variance over a single estimator.



Wisdom of the crowd: ensemble classifier error

Simple example

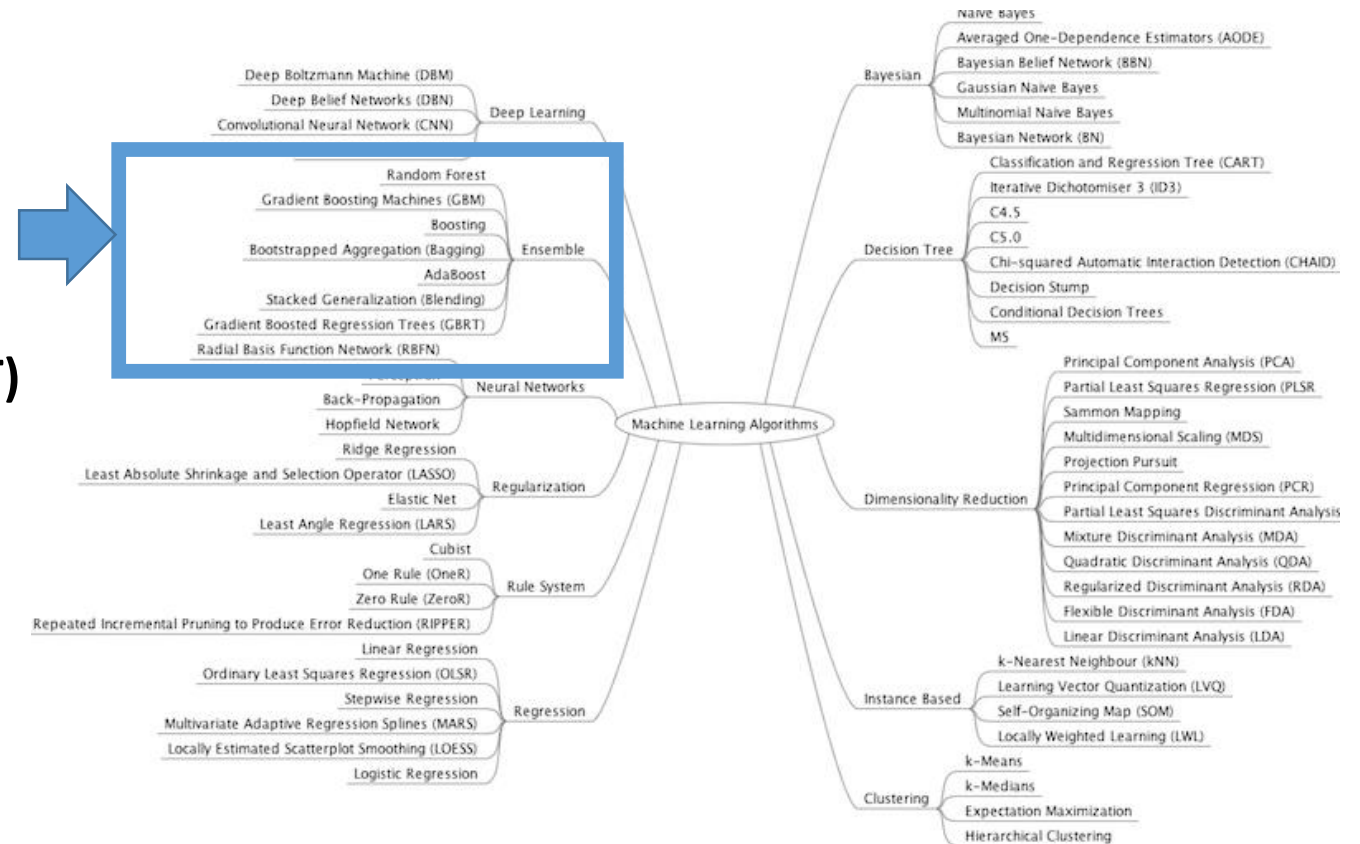
- Assuming independent classifiers
- Classifier has an error rate $\varepsilon < 0.5$
 - Ensemble prediction better than random guess
- If $\varepsilon > 0.5$ for each classifier, ensemble wrong prediction will increase



Raschka, Sebastian. *Python Machine Learning* (p. 202). Packt Publishing.

Wisdom of the crowd :ensemble learning family

- **Boosting**
- **Bootstrapped Aggregation (Bagging)**
- **AdaBoost**
- **Stacked Generalization (blending)**
- **Gradient Boosted Regression Trees (GBRT)**
- **Random Forest**



<https://machinelearningmastery.com/a-tour-of-machine-learning-algorithms/>

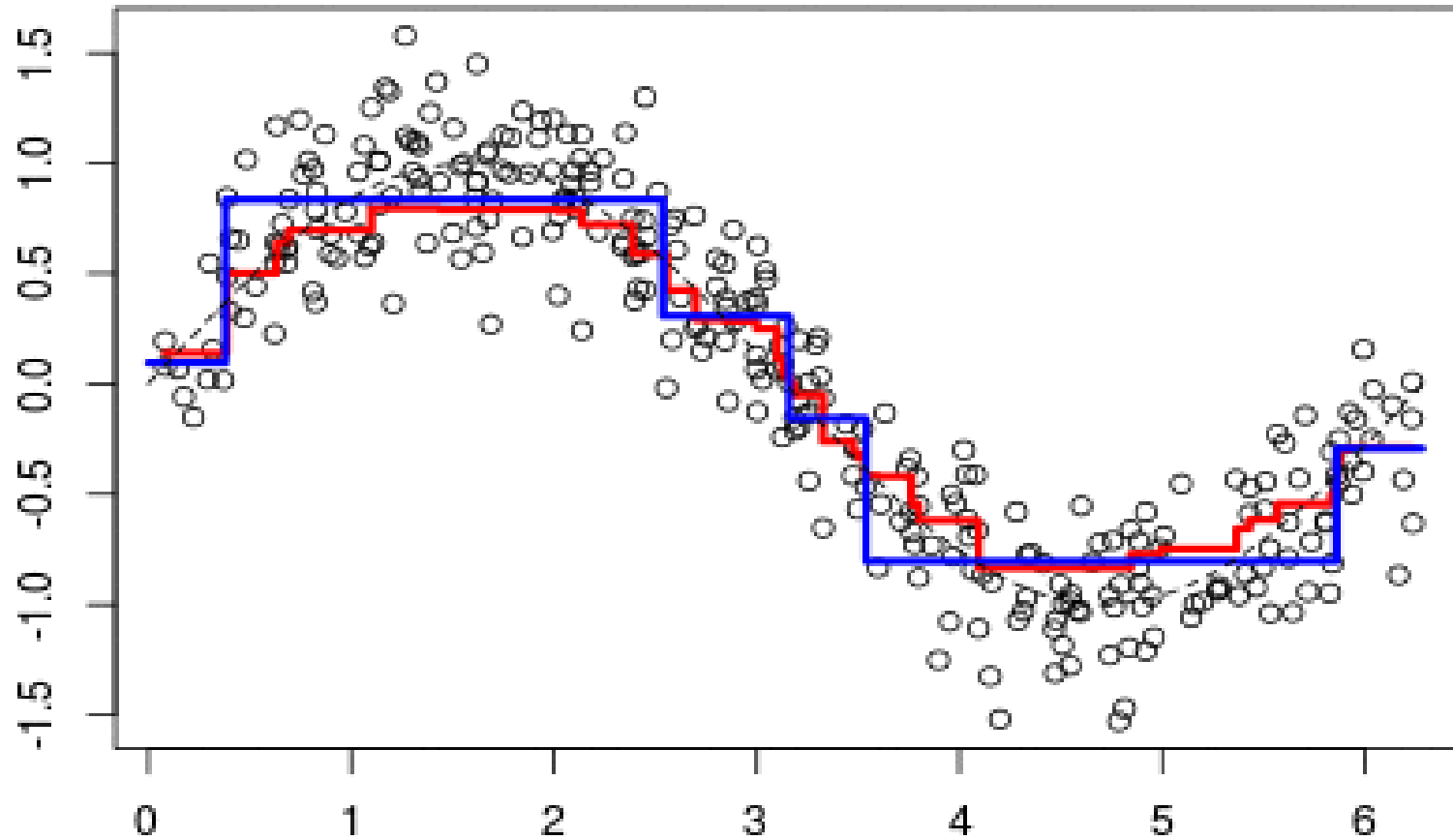
Wisdom of the crowd :ensemble learning family

Bagging: short for bootstrap aggregating is a way to decrease the variance of your prediction by generating bootstrapped samples. Each bootstrap sample is then used to fit a classifier.

Boosting: multi-step approach, where one first uses subsets of the original data to produce a series of averagely performing models and then "boosts" their performance by combining them together. Every new subsets contains the elements that were (likely to be) misclassified by previous models.

Stacking is a way of combining multiple models as inputs, that introduces the concept of a meta learner.

Bias vs Variance tradeoff



Training Equity Models: Naive example

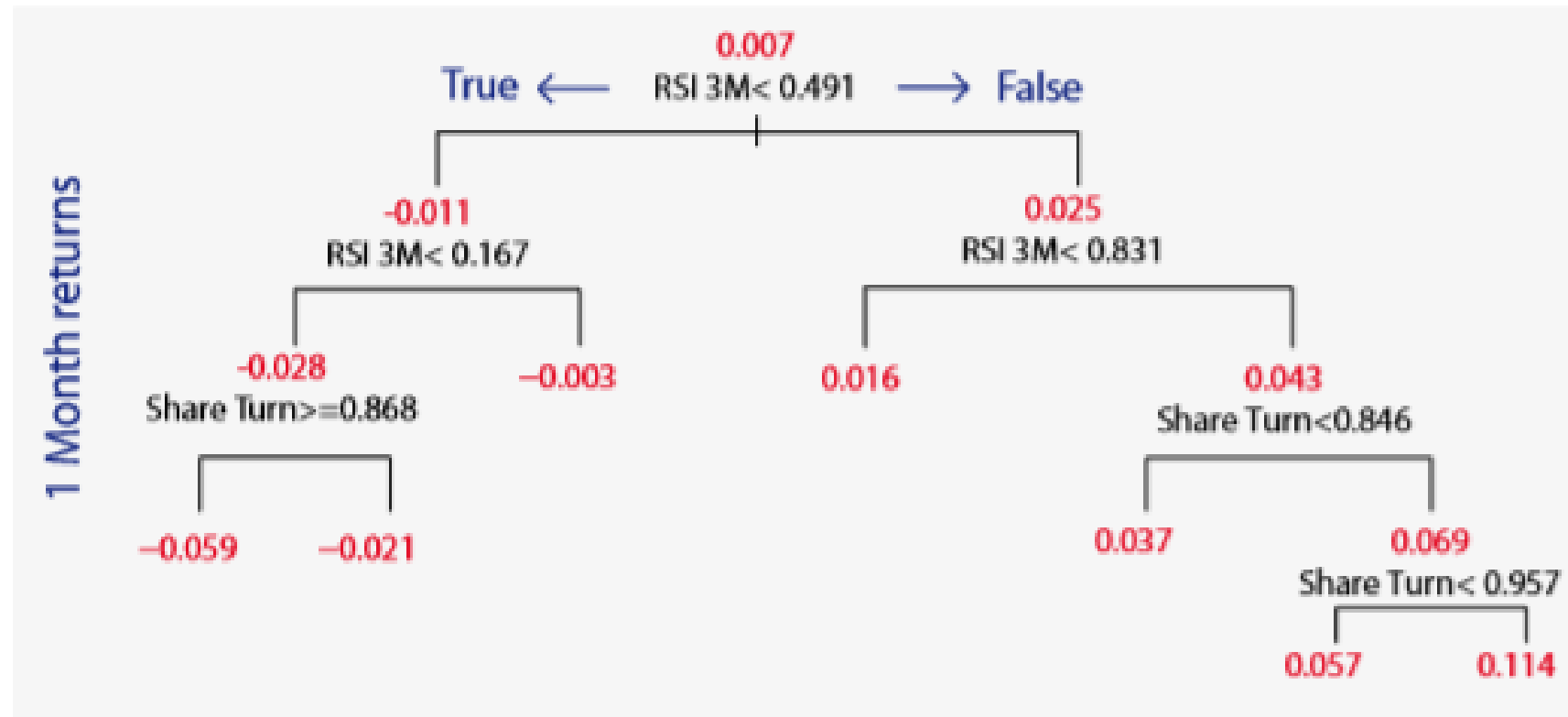
Naive approach: Objective, data and protocol

- **1M future absolute** performance
- Using simple regression trees
- Investment universe is **US stocks (~500)**
- **Full** dataset from Dec-2000 until Dec-2016
- **(~ 30) features**, monthly normalised.
- No pre-processing except winsorizing

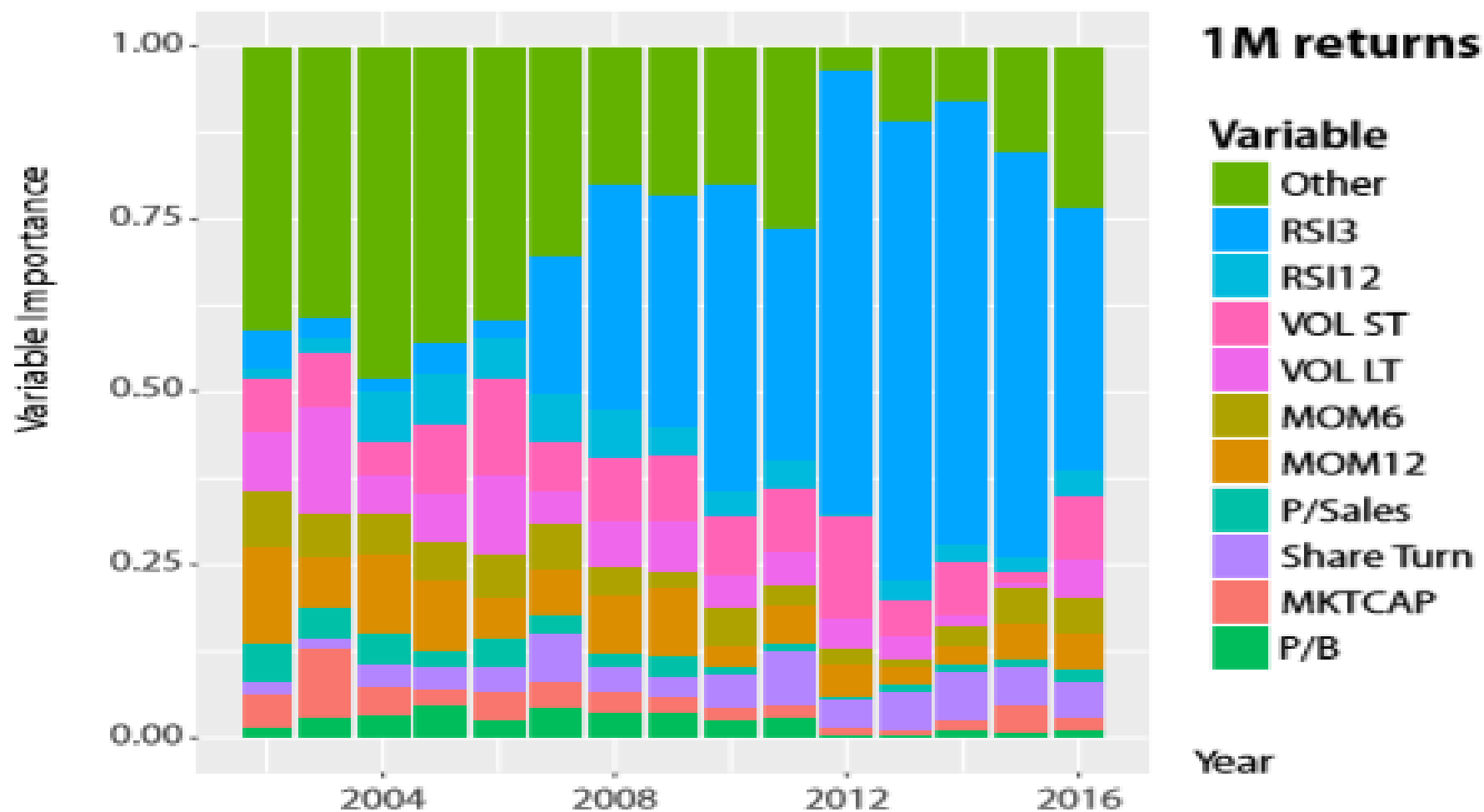
Creating the dataset

K	Short Name	Description
Fundamental / Accounting		
1	BuyBackYld	Buyback Yield
2	Cash Conversion	Cash to EBIT ratio
3	Debt/Equity	Debt on Equity
4	DivYld	Dividend Yield
5	EV/EBITDA	Enterprise Value to EBITDA ratio
6	FCFYld	Free Cash Flow Yield using FCF per Share
7	GrProfitGrowth	Gross Profits Growth (12 months)
8	Net Debt	Net Debt
9	NetDbtYld	Net Debt Yield
10	Net Prt Margin	Net Profit Margin
11	Op Prt Margin	Operating Profit Margin
12	P/B	Price to Book ratio
13	P/E	Price-Earnings ratio
14	P/Sales	Price to Sales ratio
15	PyoutRatio	Payout Ratio (Divided Per Share / Earnings Per Share)
16	ROC	Return On Capital
17	ROE	Return On Equity
18	ShareHldrYld	Total Shareholder Yield
Momentum		
19	MOM6	Momentum 6-1 months
20	MOM12	Momentum 12-1 months
21	RSI 3	RSI over 3 months
22	RSI 12	Relative Strength Index over 12 months using weekly returns
Volatility		
23	Vol LT	Ex-post volatility over 156 weeks
24	Vol ST	Ex-post volatility over 52 weeks
Liquidity		
25	ADV20	Average Daily Volume over 20 days
26	ADV60	Average Daily Volume over 60 days
27	ADV12M	Average Daily Volume over 12 months
28	BidAsk	Bid-Ask Spread
29	MKTCAP	Market Capitalization
30	Share Turn	Share turnover ratio

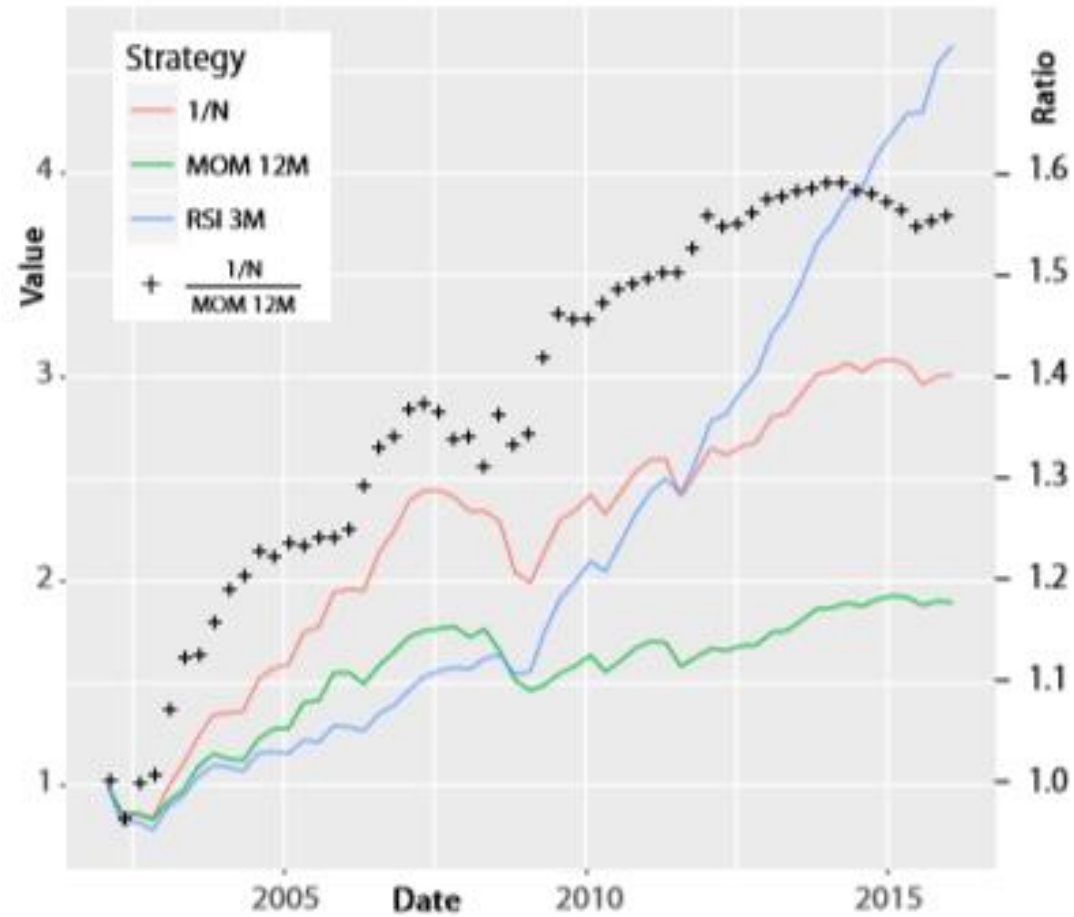
Result for the model full sample



Variable importance for the model



Implementing the model



Training Equity Models: Using a robust approach

eXtreme Gradient Boosting

XGBoost is an open-source software library which provides the gradient boosting framework. It became popular in the computer science community after being used by most of winners in kaggle competition



is a platform for predictive modelling and analytics competitions in which statisticians and data miners compete to produce the best models for predicting and describing the datasets uploaded by companies and users.

eXtreme Gradient Boosting : introduction to the model behind

General objective of tree ensemble for K trees

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F}$$

$$Obj = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

Training on loss

Complexity of the trees

$$\begin{aligned}\hat{y}_i^{(0)} &= 0 \\ \hat{y}_i^{(1)} &= f_1(x_i) = \hat{y}_i^{(0)} + f_1(x_i) \\ \hat{y}_i^{(2)} &= f_1(x_i) + f_2(x_i) = \hat{y}_i^{(1)} + f_2(x_i) \\ &\dots \\ \hat{y}_i^{(t)} &= \sum_{k=1}^t f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i).\end{aligned}$$

Additive training

<http://xgboost.readthedocs.io/en/latest/model.html#>

Why using eXtreme Gradient Boosting ?

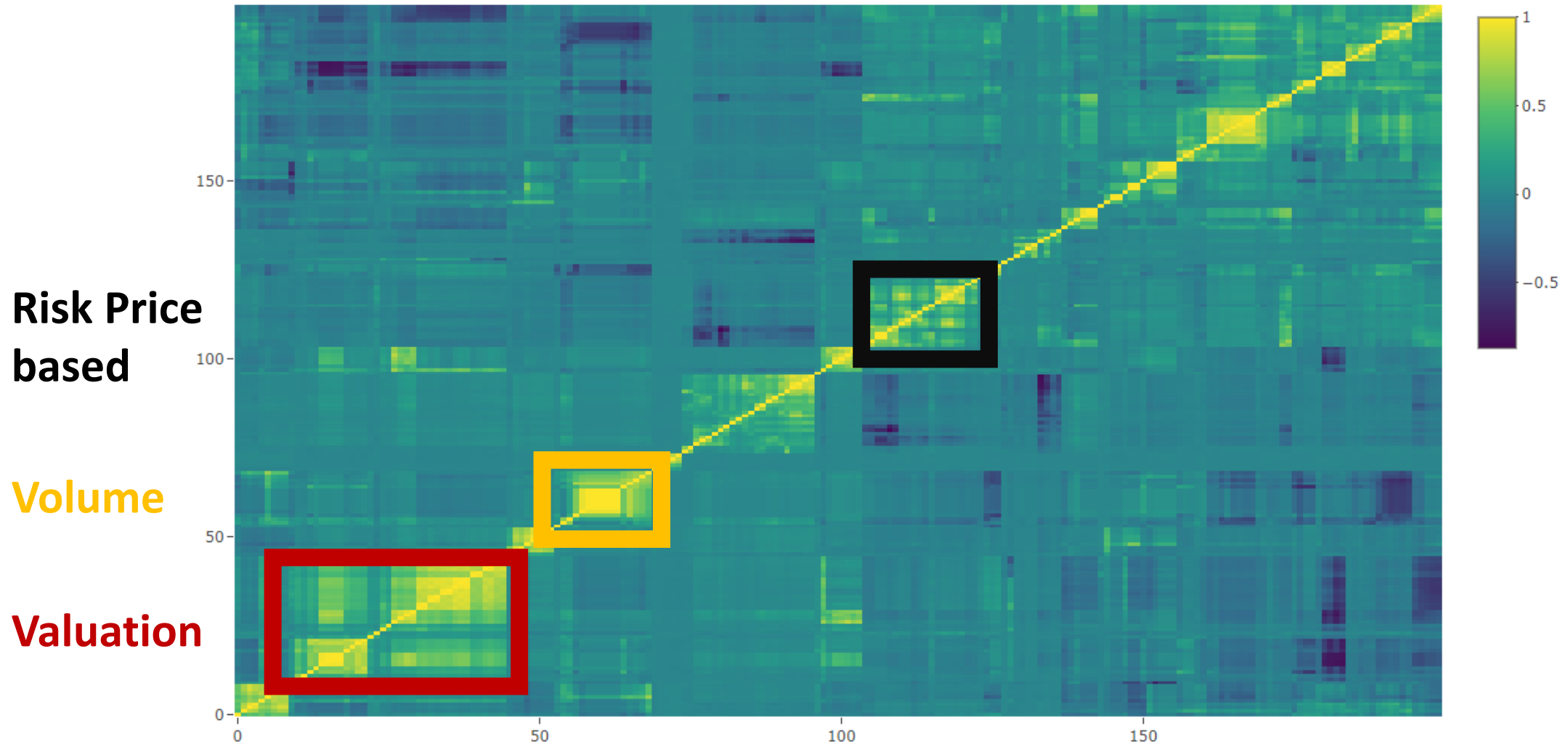
- **Easier** to detect future **outperforming** stocks using non-linearity and **variables interactions**
- Wisdom of the **crowd**
- **Model complexity controlled** through **regularization parameters**
- Computation speed, **parallelized**
- Interpretability through **variable importance**

XGBoost applied to Quant Equity

Objective, data and protocol

- We will try to predict **12M future** performance relative to their sectors.
- Investment universe is **US stocks (~3000)**
- **Full** dataset from Dec-2000 until Dec-2017
- **(~ 200) features**, monthly normalised in percentrank.
- Dataset pre-processed, outliers removed, focussing on training on the tails of the distribution (**top and bottom 20%**)
- Split the dataset between **Training (80%)** and **Testing (20%)**
- We will use **k-fold cross validation to find hyperparameters**
- Rolling window of **24 months**

Features discovery : « brute force » approach



Hyperparameters used for tuning

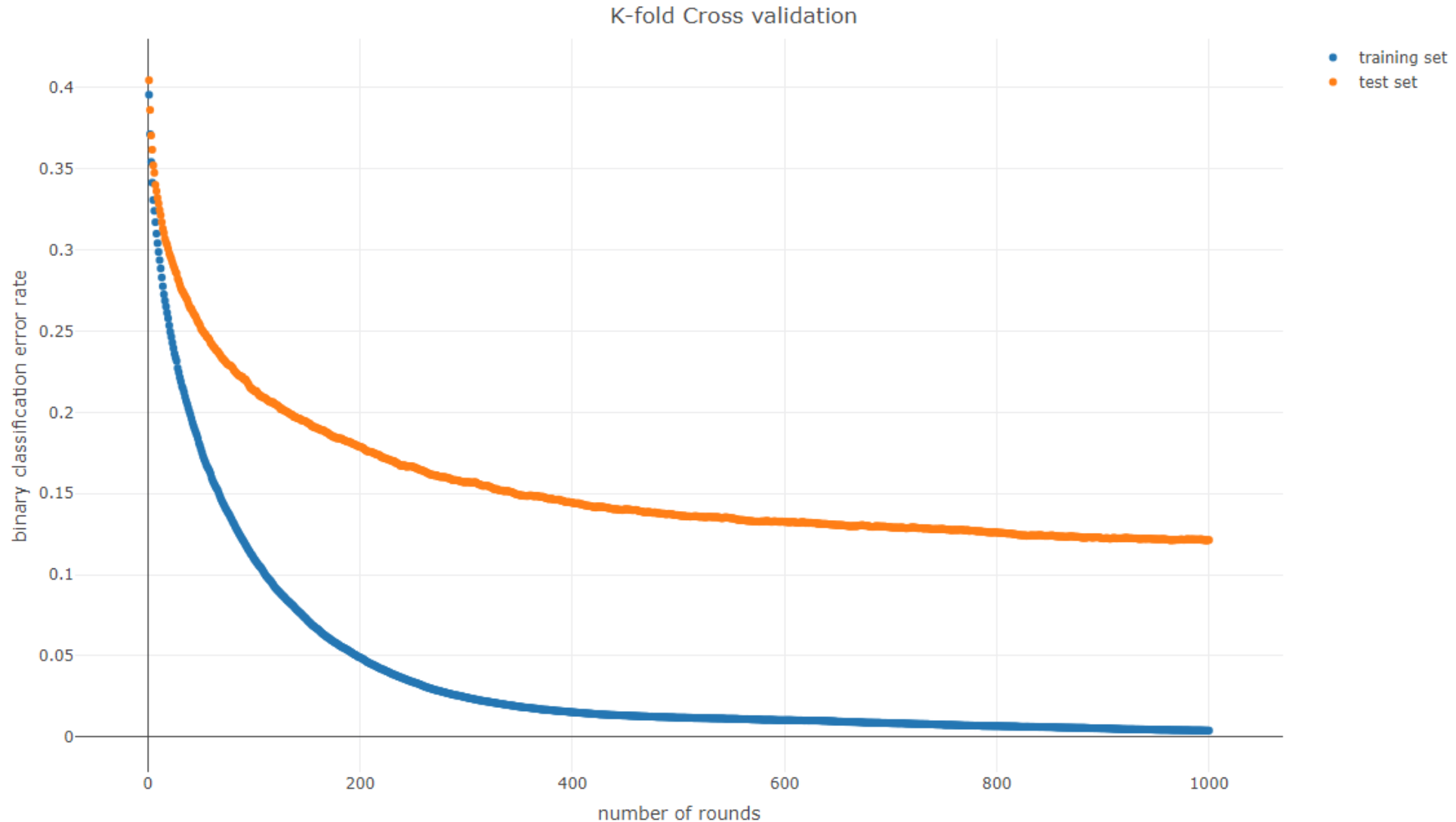
learning_rate : step size shrinkage used in update to prevents overfitting. After each boosting step, we can directly get the weights of new features. and eta actually shrinks the feature weights to make the boosting process more conservative.

min_split_loss : minimum loss reduction required to make a further partition on a leaf node of the tree. The larger, the more conservative the algorithm will be.

max_depth : maximum depth of a tree, increase this value will make the model more complex / likely to be overfitting. 0 indicates no limit, limit is required for depth-wise grow policy.

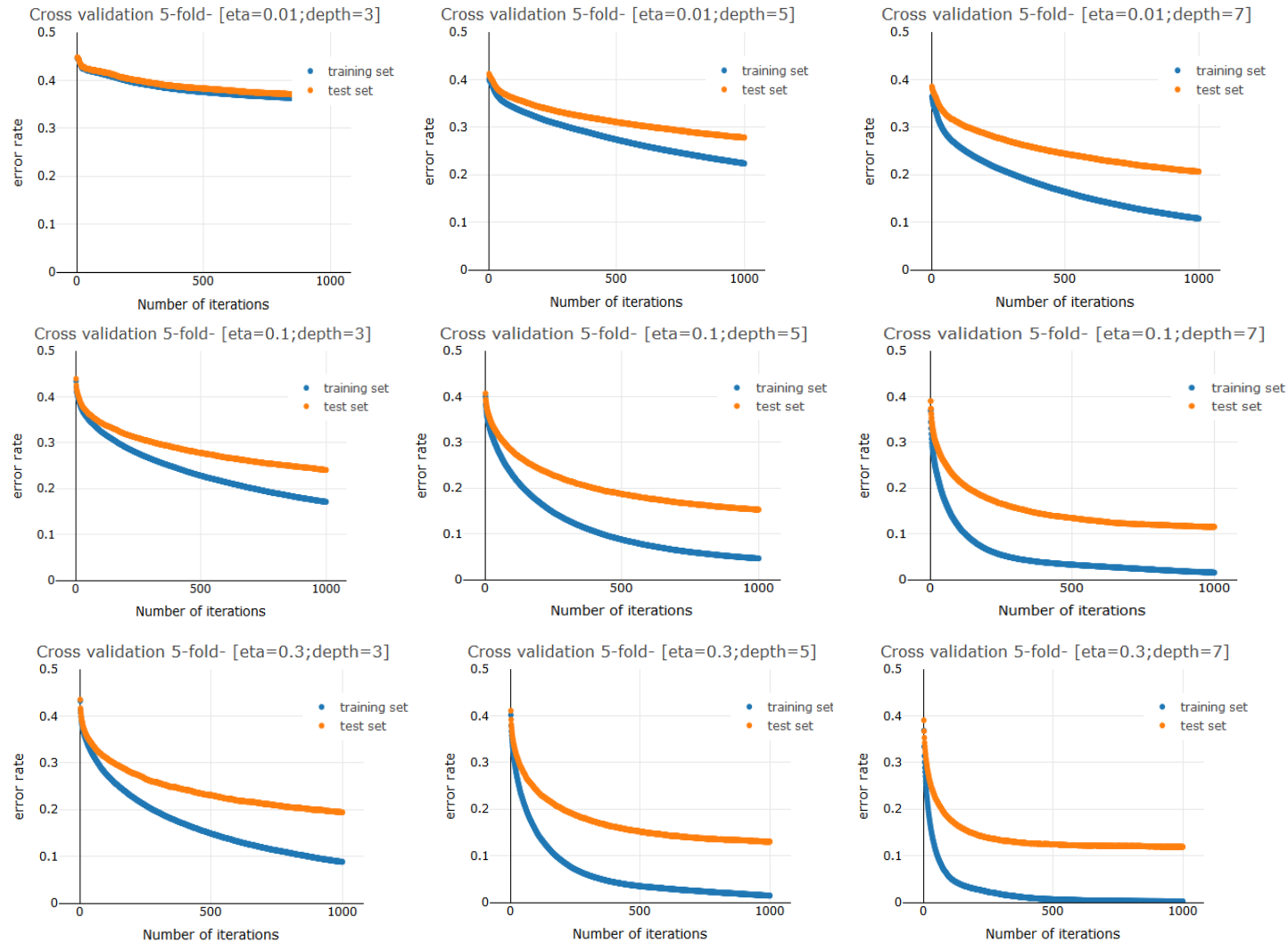
scale_pos_weight : Control the balance of positive and negative weights, useful for unbalanced classes.

Cross validation: choosing the hyperparameters



Max_depth=5, eta=0.1 gamma=0.1, scale_pos_weight =sum_pos/sum_neg

Cross validation: choosing the hyperparameters



type I type II errors

	OUTPERFORMED	UNDERPERFORMED
OUTERPERFORM	True Positive: Stock WAS classified as outperforming sector and DID outperformed	False negative: Stock was NOT classified as outperforming sector and DID outperformed
UNDERPERF.	False Positive: Stock WAS classified as outperforming sector and did NOT outperformed	True Negative: Stock was NOT classified as outperforming sector and did not outperformed

- Left side of the matrix shows **Actual**
- Top side shows **predicted**

Class error 29%

Class error 25%

Error Rate
27%

Beyond type I type II errors

- **Fp** : false positive. Stock predicted to outperform and that did not outperform out of sample.
- **Fn** : false negative. Stock predicted to underperform that outperform out of sample.
- **Tp**: true positive. Stock predicted to outperform which outperform out of sample.
- **Tn**: true negative. Stock predicted to underperform which underperform out of sample.

Precision: $Tp / (Tp + Fp)$

Precision could be defined as a rate of successful prediction for sector neutral outperforming stocks.

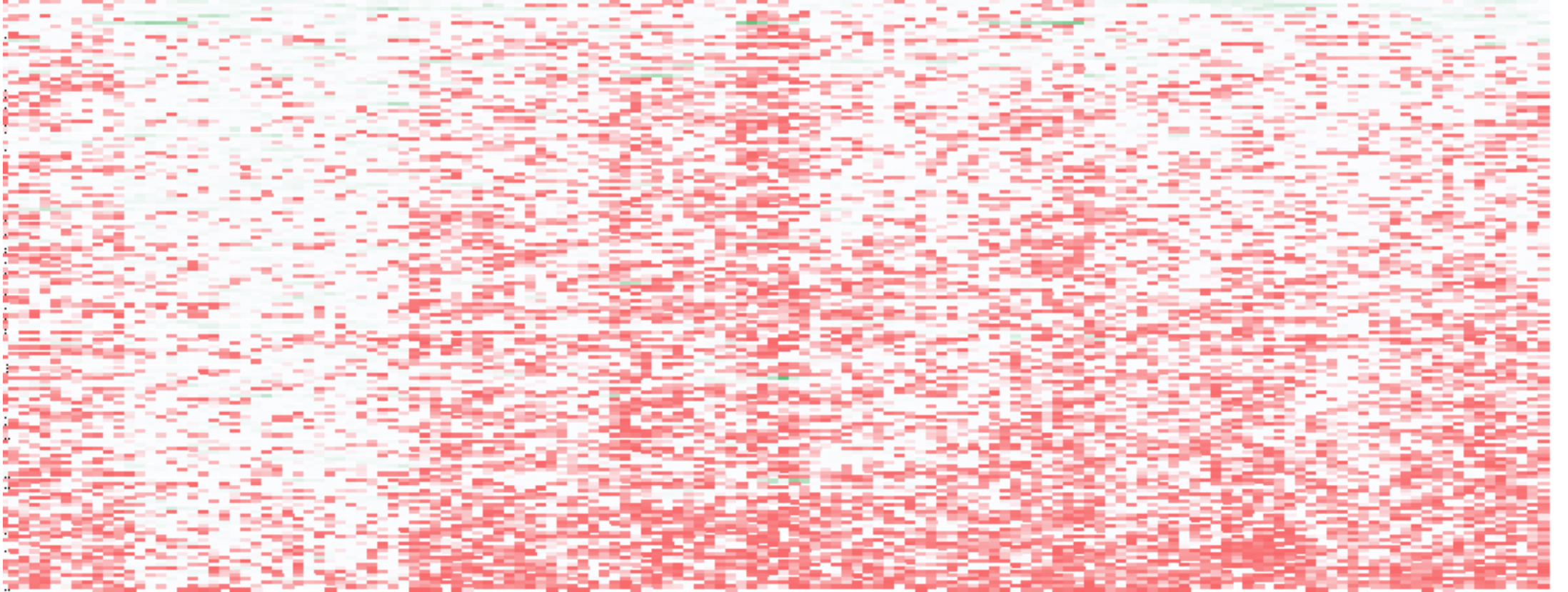
Recall: $Tp / (Tp + Fn)$

Recall could be defined as a true rate, since we include the instances that have been wrongly classified in negative.

Accuracy: $(Tp + Tf) / (Tp + Tn + Fp + Fn)$

This is the accuracy level used in the cross validation part.

Variable importance rolling model: 100 signals left

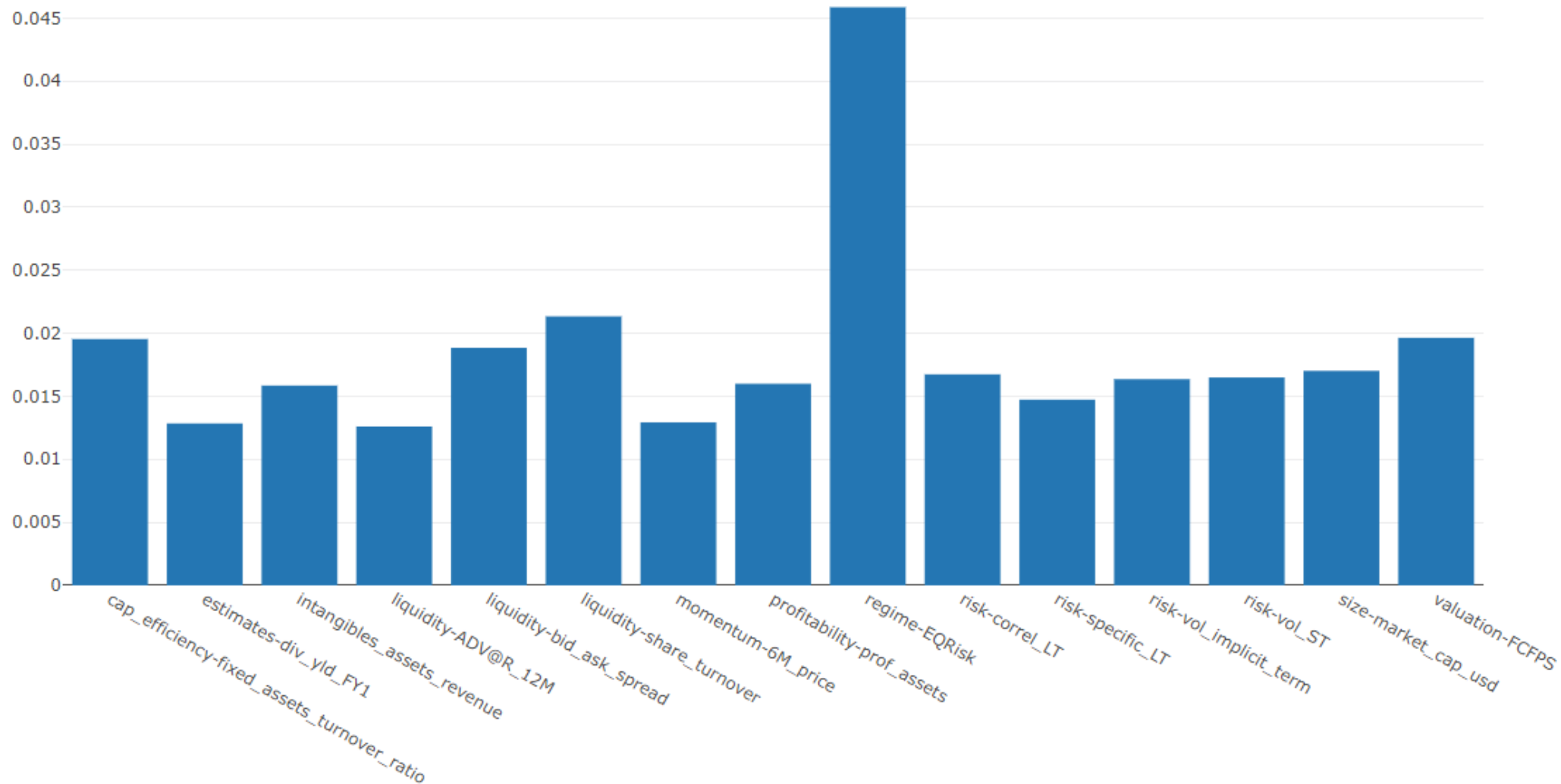


Variable importance rolling model : regime based signal



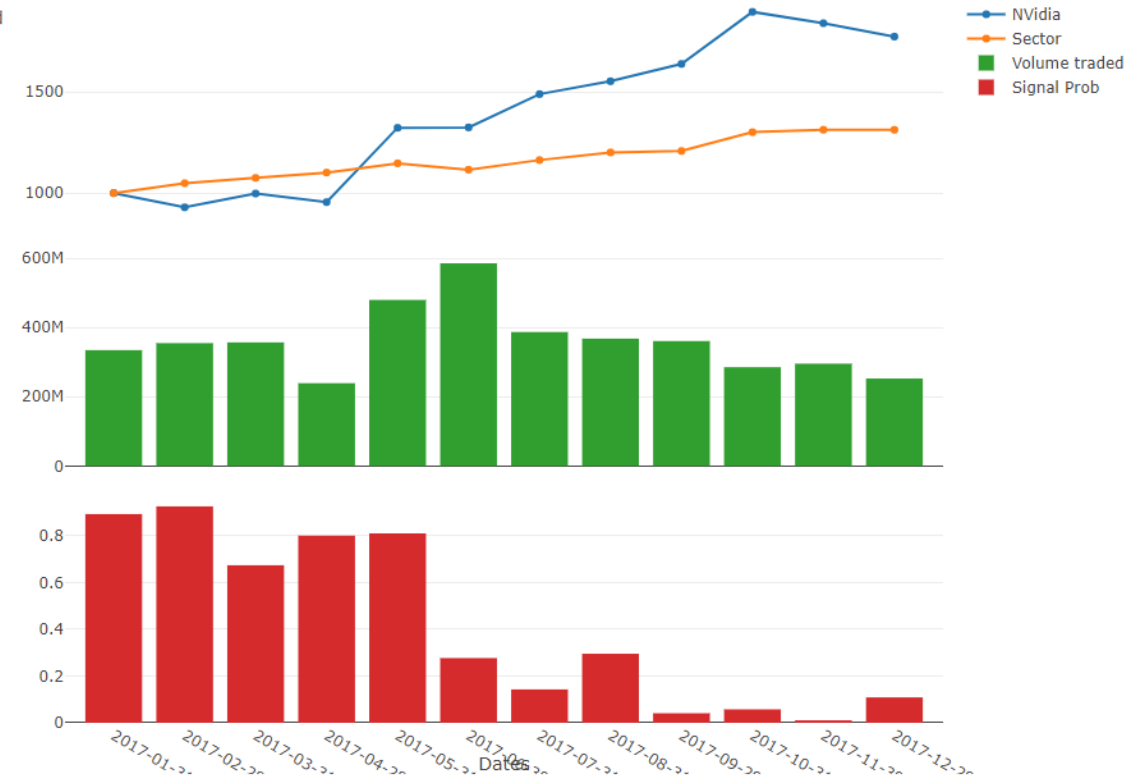
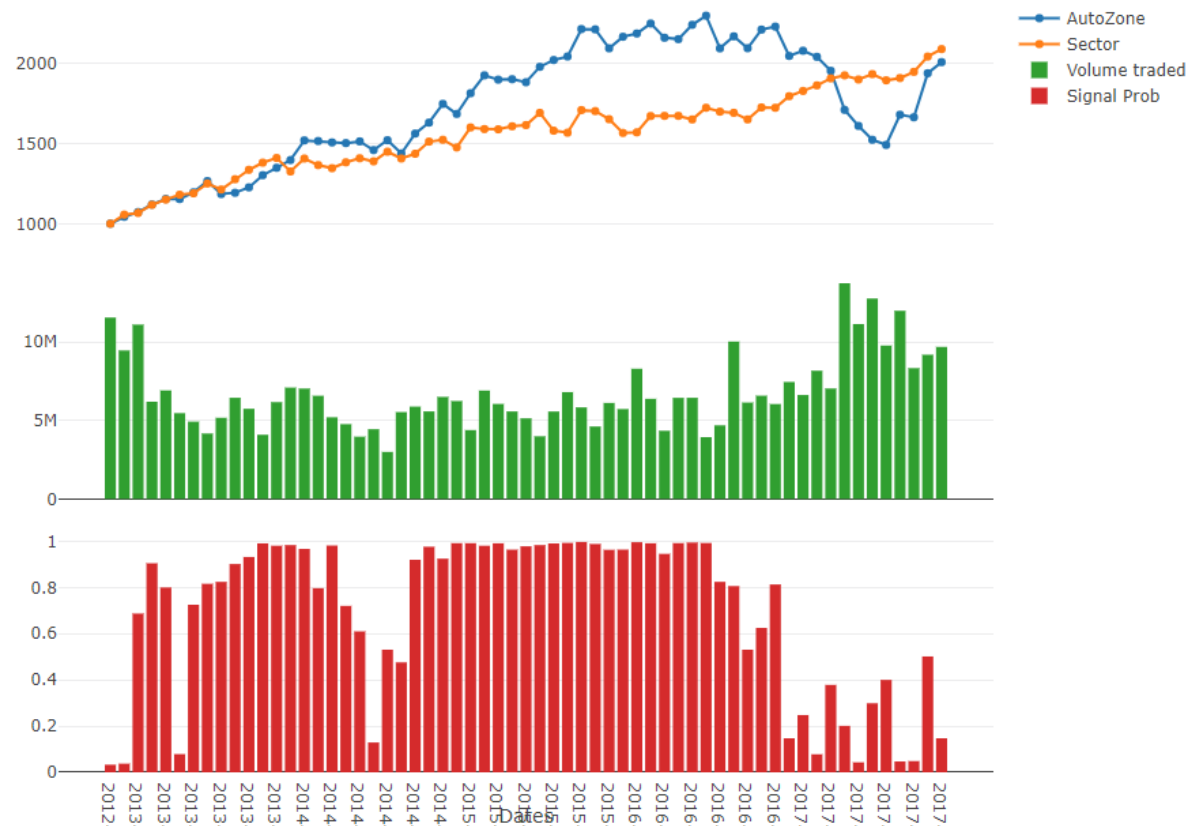
Variable importance rolling model

"Top 15 features- gain avg full sample"



Implementing in a Quant Process

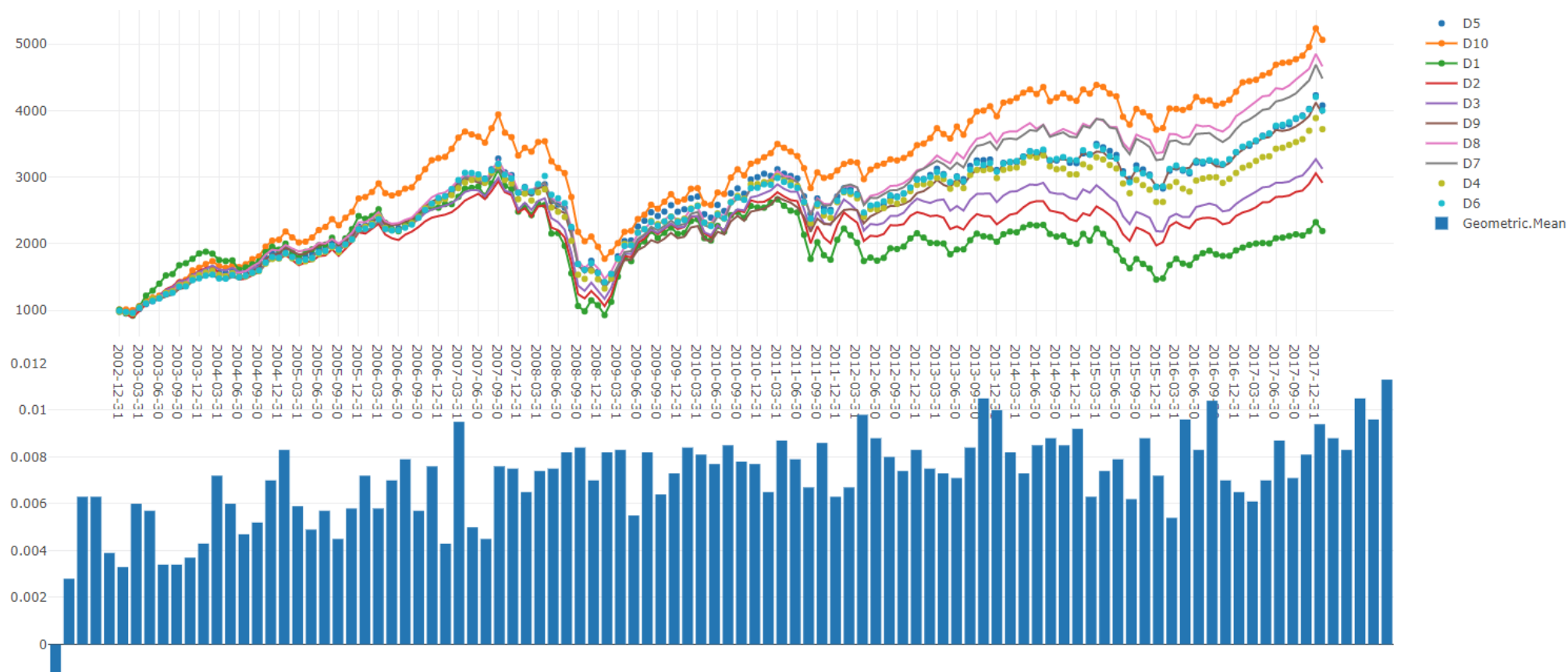
Trading single stock using the ML signal?



Implementation : ML vs traditional signal blending approach

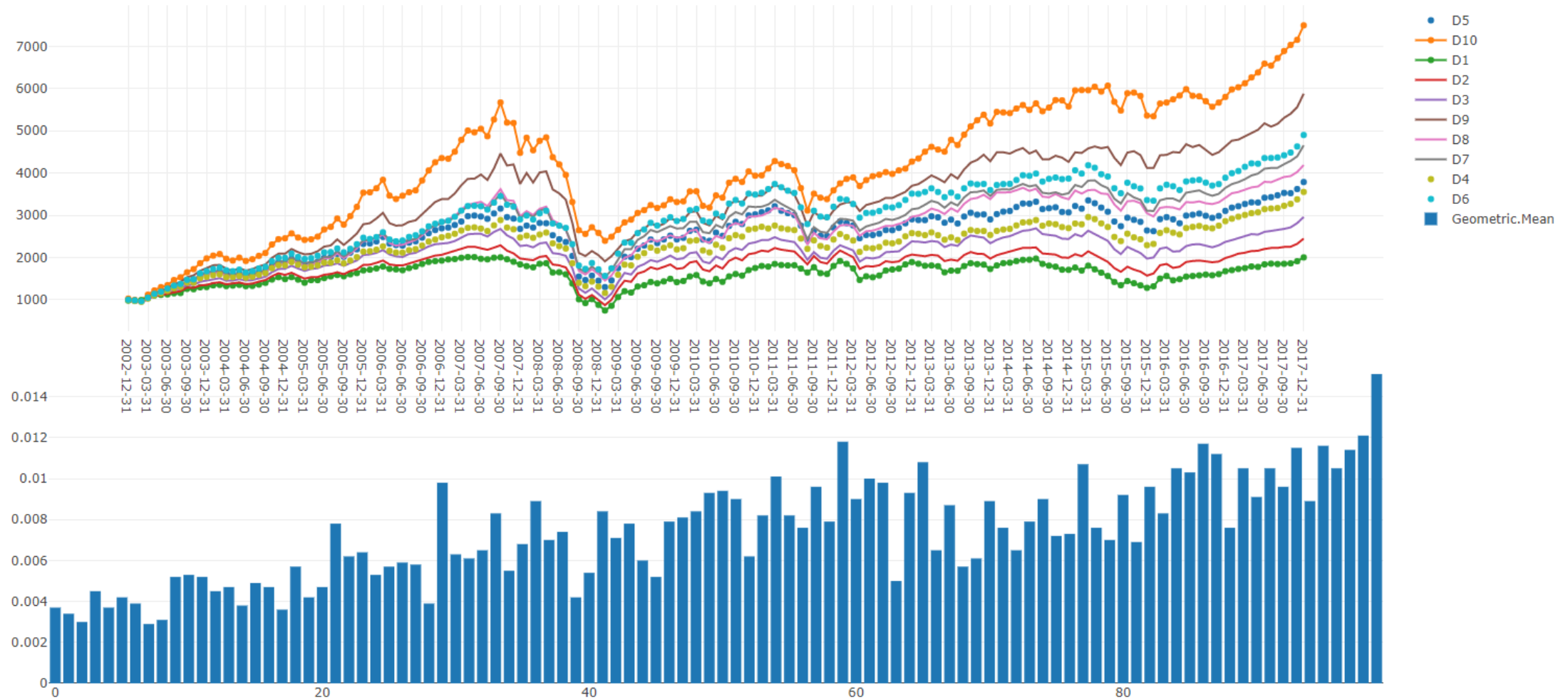
- **EW portfolio** based on the top decile of the ML signal
- Comparing against a simple linear average of composite factor signal (**Mfactor**) made of
 - **Quality** : ROE, Debt/Equity
 - **Volatility**: 3Y and 1Y (weekly returns)
 - **Value**: PE, PB, EV/EBITDA
 - **Momentum**:12-1 returns
 - **Size** : Market cap

Multi-factor signal analysis: Decile performance



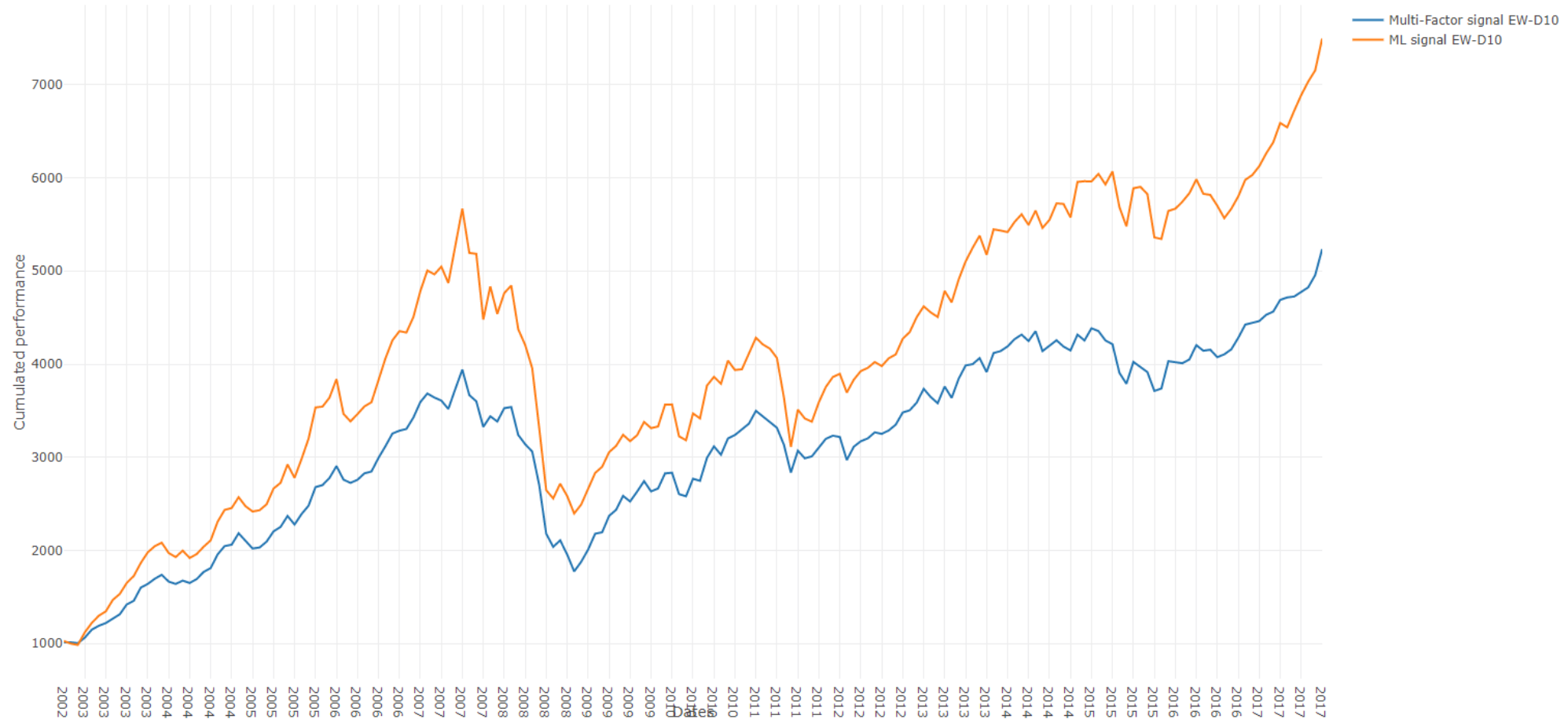
D10 portfolio: tstats =2.69, return p.a.=11.8%, ann. Vol=14.8%, Turn p.a. (2-ways)=123%, TC=62bps p.a.

ML signal analysis: Decile performance



D10 Portfolio: tstats =2.99, return p.a.=16.9%, ann. Vol=16.8%, Turn p.a. (2-ways)=192%, TC=85bps p.a.

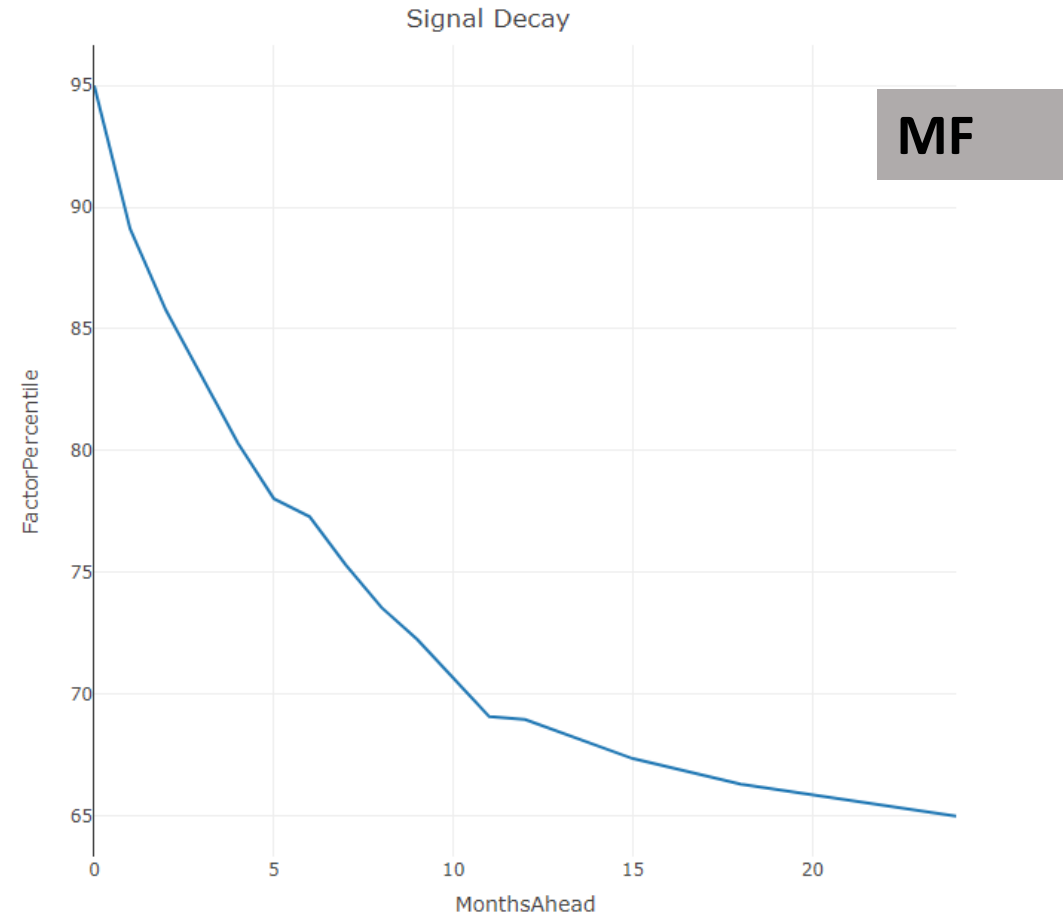
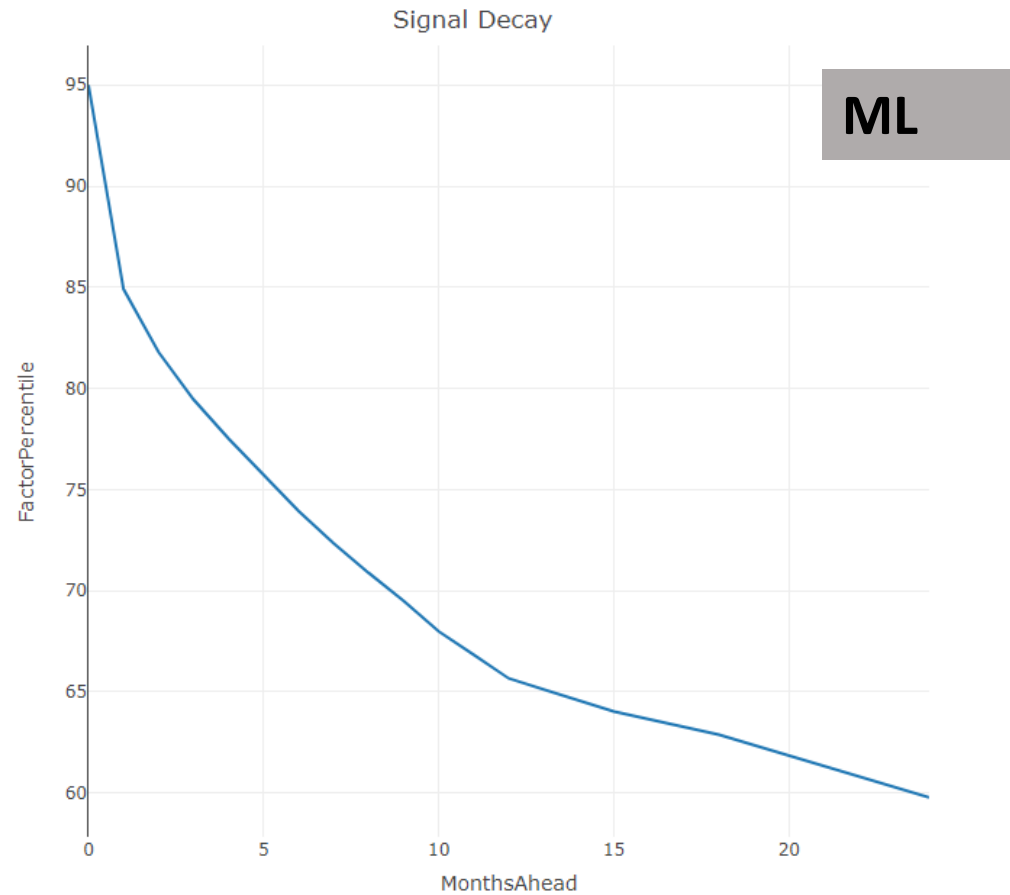
Portfolios' performance comparison



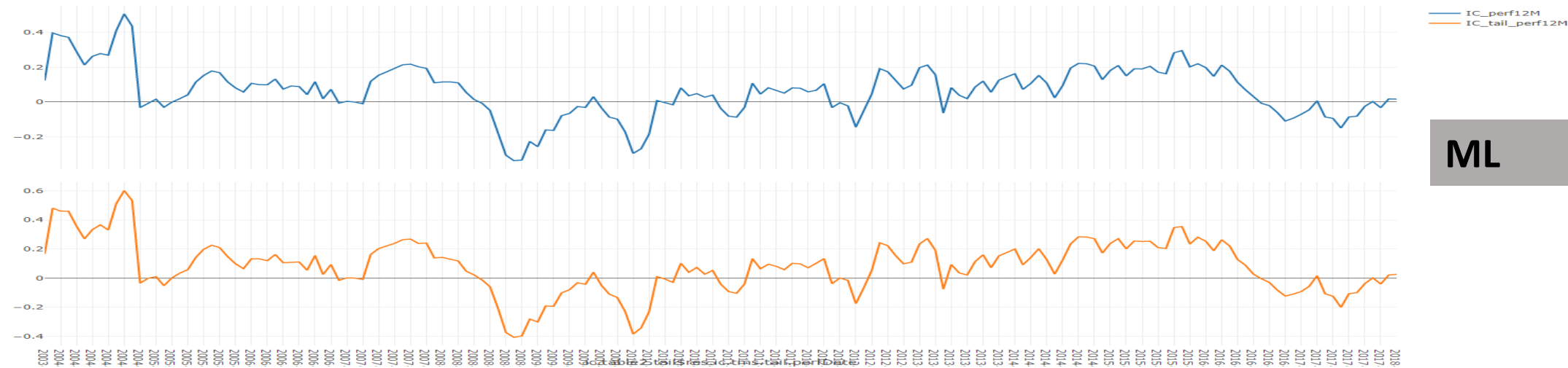
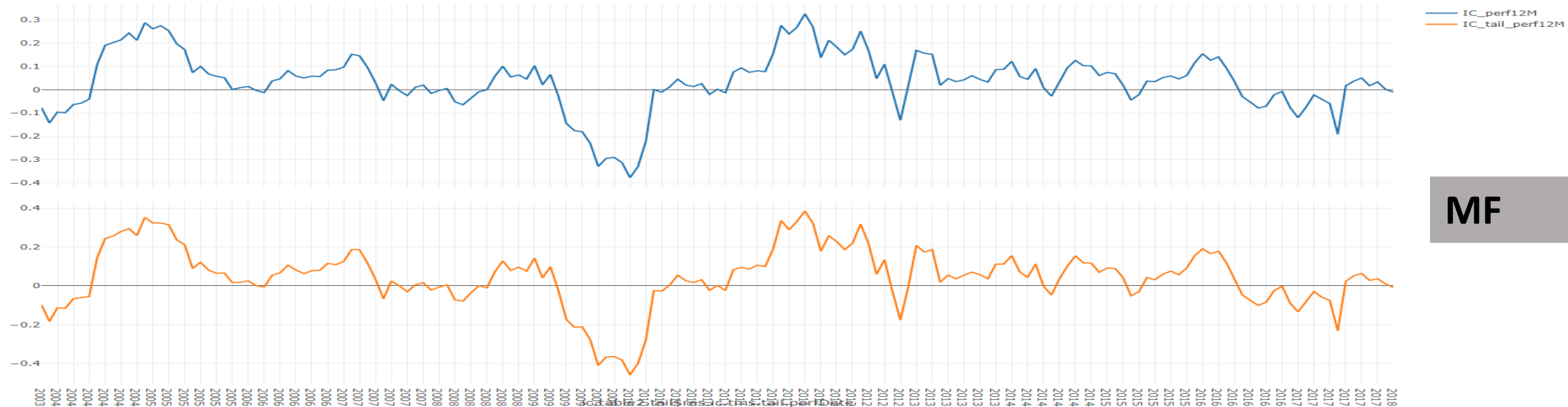
Portfolio ML: tstats =2.99, return p.a.=16.9%, ann. Vol=16.8%, Turnover p.a. (2-ways)=192%

Portfolio MF: tstats =2.69, return p.a.=11.8%, ann. Vol=14.8%, Turnover p.a. (2-ways)=123%

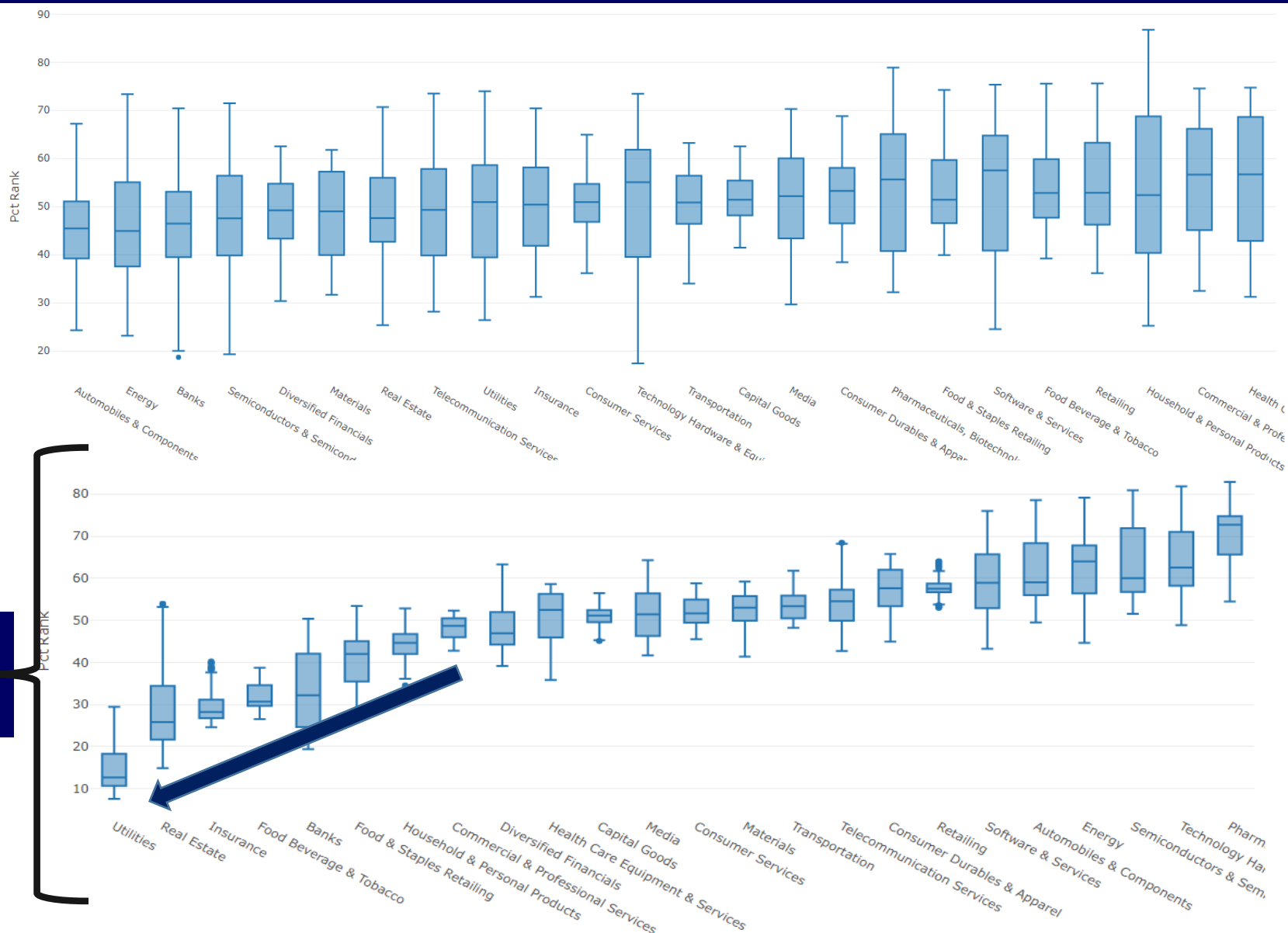
Analysing signal's decay



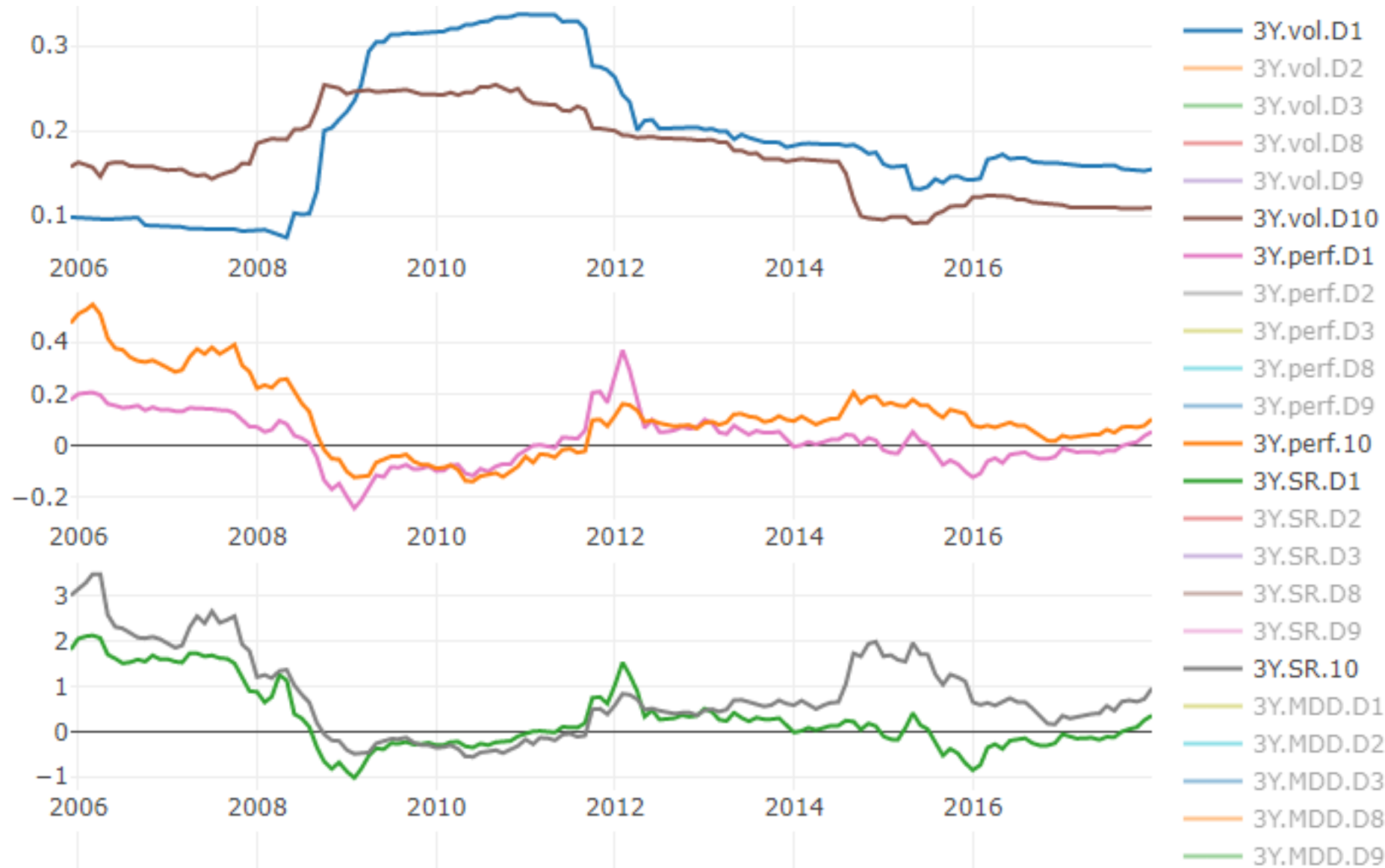
Information coefficient



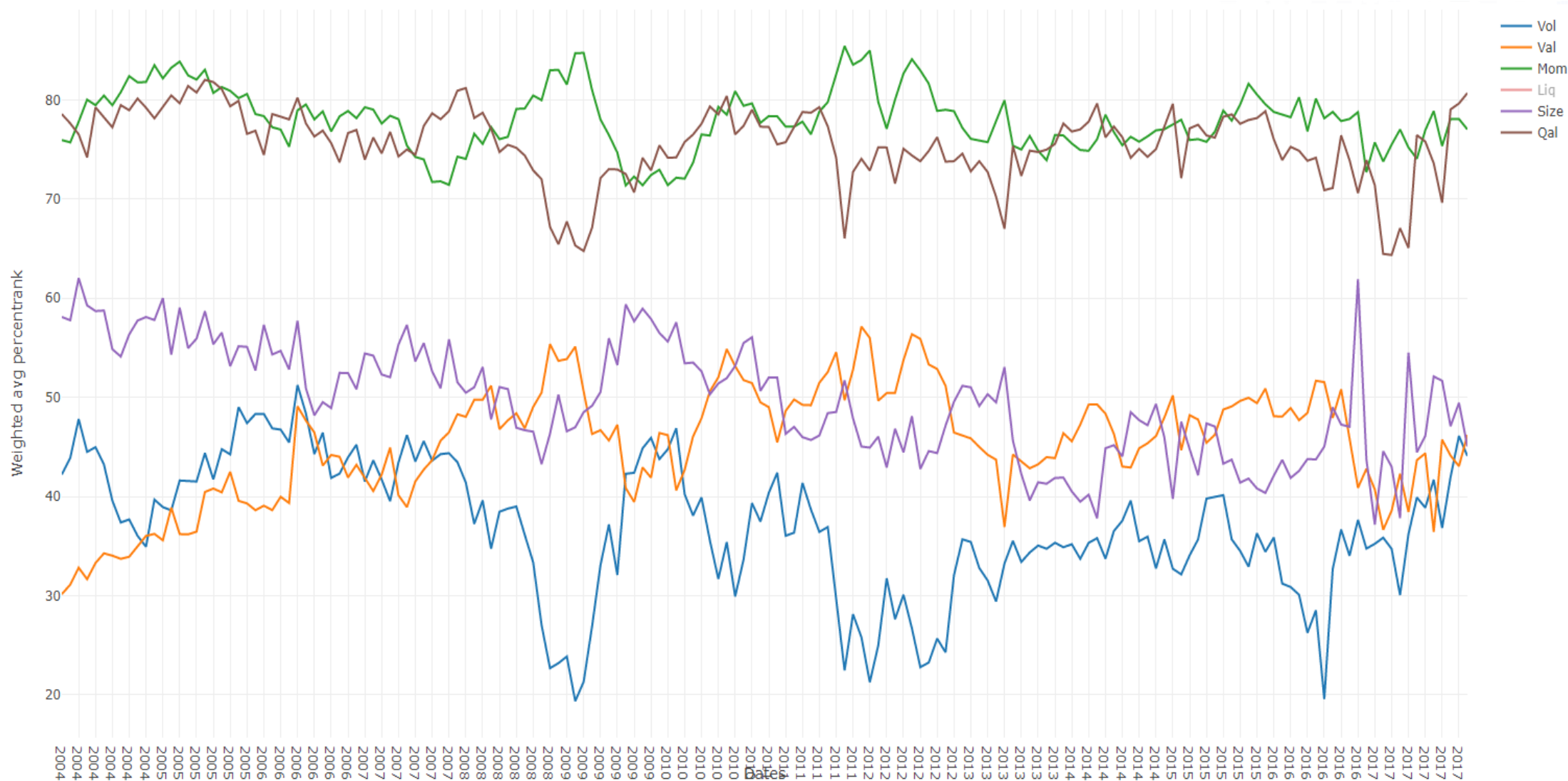
Comparing potential sector bias



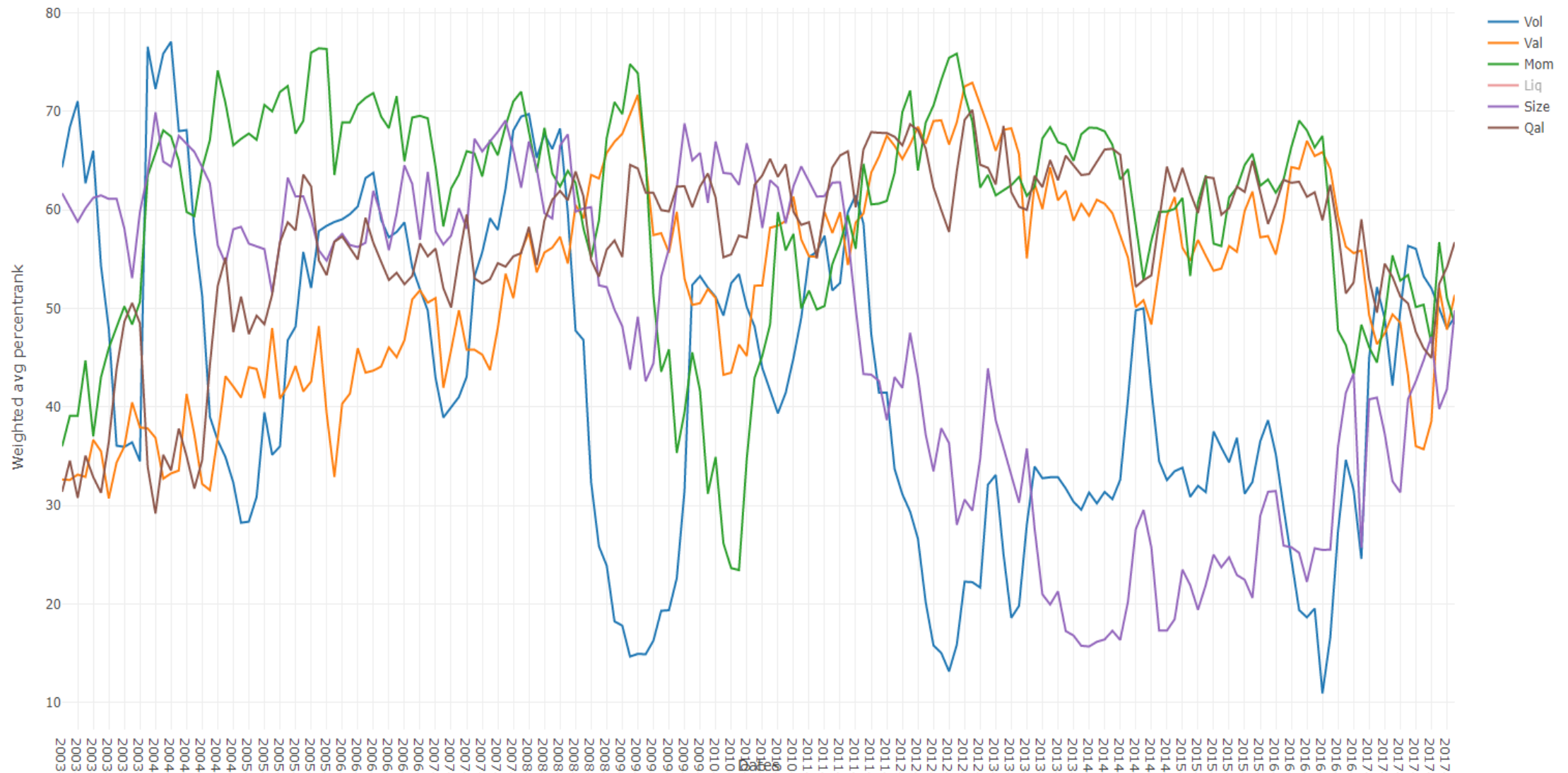
3Y Rolling analytics for EW portfolios based on ML signal



Holdings-based factor exposure: multi factor based portfolio (D10)



Holdings-based factor exposure: ML based portfolio (D10)



Conclusion

- ML in finance could not be applied as a pure copy paste from computer science.
- Yet, when properly addressed, ML algorithms could be extremely useful in any Quant process.
- By applying features engineering on core financial data and using more « orthogonal » datasets (NLP based, alternative data..) ML based modelling has a huge potential for improvement.
- Embracing modern approaches like ML and alternative/big data will be soon a matter of “survival” for innovation.

Disclaimer & source

The views expressed in this document are the author's and do not necessarily reflect those of the organizations he is affiliated with.

No investment decision or particular course of action is recommended by this presentation from the author

Source: QuanDL, Alphavantage, R & Rstudio, Python & Spyder, XGBOOST,